Validation protocol reverses machine-learning model rankings in tokamak energy-confinement scaling

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Code, results, and reproducibility materials: `https://github.com/lsalcedo100/FusionFlux`
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Abstract

Confinement scaling laws set the size a next-step tokamak has to reach. On the ITPA Global H-mode Confinement Database (HDB5, STD5; 6228 slices, 18 labels), under cross-validation grouped by discharge, a random forest cuts log error by 29% against a power law refitted in the same folds and 36% against published IPB98(y,2). That margin does not survive the question a scaling law exists to answer: held out one device at a time, the ordering of the three primary models reverses. Two controls set its strength. The loss power is the target's denominator, and regressing the stored energy instead leaves the forest worse on **9 of 13** labels and 9 of 11 devices at a mean gap of +0.220, neither exact test reaching $p = 0.05$. Several features barely move within a device; in dimensionless coordinates, where none is a device label, it is 11 of 13 and 9 of 11. On the confinement time and engineering features it is 13 of 13 and 11 of 11; those two effects are the difference. It also survives nested tuning, threshold and fold sweeps, and equal label weighting. The forest's error tracks the held-out label's Mahalanobis distance from the training data at $\rho = +0.85$, against -0.06 for the power law and $+0.56$ for a constant baseline. At a cut reproducing ITER's 1.82× jump in major radius the trees land nearer a constant predictor than the power law, and 90% split-conformal intervals fall to 3% and 0% coverage. Two repairs help, both supplying long-range structure rather than capacity: Connor–Taylor constraints, whose Kadomtsev rung is free in sample, and a Gaussian process with a linear kernel component. That process is this work's best cross-validated model and does not reverse. Extrapolation failure tracks long-range saturation rather than flexibility.

Keywords: tokamak energy confinement; confinement scaling; machine learning; extrapolation; cross-validation; dimensional similarity; uncertainty quantification; ITER

1 Introduction

A tokamak ignites when the alpha heating from its own fusion reactions covers everything the plasma loses. The energy confinement time τ_E sets that loss rate, which makes it a

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principal input to how large a next-step device has to be. Empirical scaling remains a principal tool for projecting it to a device that does not exist yet [1]: a power law in the engineering parameters,

$$\tau_E = C\, I_p^{a_1} B_t^{a_2} \bar{n}_e^{a_3} P^{a_4} R^{a_5} \epsilon^{a_6} \kappa^{a_7} M^{a_8}, \tag{1}$$

fitted by least squares across discharges from the multi-machine confinement database. In log space Eq. (1) is exactly a linear model, so fitting one is ordinary least squares on a log design matrix. The reference fit is IPB98(y,2) [1], a regression over the ITPA multi-machine database whose exponents were used to size ITER; that database is still maintained and republished [11], and the confinement and transport basis was revised in 2007 with IPB98(y,2) retained as the reference H-mode scaling [2].

What such a law is used for is an extrapolation, and a large one. ITER's major radius is 1.82 times the largest machine in the database, and SPARC [12] is designed for a toroidal field 2.1 times the largest recorded in it. The law is asked for a number at a point no row occupies, and that is the regime in which a fitted regression carries no guarantee at all.

Empirical scaling is not the only route to that number, and the field's answer to the difficulty just described has been to stop relying on it alone. First-principles and reduced transport models project confinement from the turbulence that sets it rather than from a regression over machines already built, and the integrated modelling assembled around them is a standard part of how a next-step device's performance is now estimated [2]. The machine learning in that branch is aimed at a different target from the one here: neural surrogates are trained to reproduce a physics-based transport model fast enough to run inside a simulation [3, 4], so what is being regressed is a solver rather than a database of machines. SPARC's core performance is projected that way [5] alongside the empirical route. The two routes fail in unrelated ways, which is most of why both are used, and nothing below is a claim about the second or evidence against it. What is asked here is narrower and concerns the empirical branch on its own terms. It is still fitted, still published, still quoted as a design point and still used to benchmark the physics-based projections, and it is the practice that selects among empirical models that turns out not to measure what it is used for.

This paper reports two things about that regime, and both are worth separating from the expectation that accuracy simply degrades outside the data. The first is that the *ranking* of candidate models reverses. Under cross-validation grouped by discharge a random forest beats a power law refitted in the same folds by 29%; held out one whole device at a time, it is worse than that same power law on every device scored. A practitioner following the conventional procedure does not merely get an optimistic score for the right model, they get the wrong model. The second is what the calibrated intervals do *instead of* widening. Split-conformal intervals set to 90% cover 3% and 0% of held-out rows at a size cut matched to ITER's, and their width moves by under 1.5% between the two arms. The stability of the width is not itself a discovery: a split-conformal half-width is a rank statistic of the calibration residuals and cannot depend on the test distribution at all. The consequence is what is being reported. A design study handed such an interval cannot tell from the interval that it has failed, because the one quantity that would carry the warning is fixed by the construction rather than by the data it is applied to. Degradation is a matter of degree and can be absorbed by a margin. Neither of these can.

A third observation follows from the same table and points at what to do about it. On a held-out label the published IPB98(y,2) law scores 0.188 against 0.214 for the power law refitted here on the same rows and features, and 0.194 against 0.278 at the size cut. Counted the way every other comparison in this paper is counted, it wins **9 of the 13** labels outright, at an exact $p = 0.27$ that establishes nothing on its own and a direction that is nonetheless the same one the means give. A law fitted in 1998 to a predecessor database transfers better than an unconstrained refit of the same functional form on more data. Which rows each saw is part of that and is not all of it. IPB98(y,2) was fitted under an enforced high-β Kadomtsev constraint with the field exponent held at 0.15 [1, 11], which is exactly the long-range structure Sec. 8 finds to be worth having: imposing the same family of constraints on a refit here recovers part of the gap without costing anything in sample, and Sec. 8 reports how much of it, by rung, together with where the newer published laws sit on the same surfaces, which is not where their scores would predict. The constructive half of this paper follows from that, and the ranking inversion is what makes it necessary rather than optional.

Replacing Eq. (1) with a flexible regressor is an obvious move, and the practice is not new: Allen and Bishop [9] fitted a neural network to confinement data in 1992 and reported that it beat linear regression on an in-distribution score, which is the comparison this paper is about, made thirty-four years ago. Machine learning has been productive elsewhere in this field since: disruption prediction [13], transfer of a disruption predictor to a device that supplied almost none of its training data [14], and real-time magnetic control [15]. Confinement scaling itself has been revisited with symbolic regression, which abandons the power-law form rather than fitting it [16], and with a neural residual learned on top of a frozen power law [17]. Measured conventionally the flexible models win. On the rows used here the random forest scores 0.128 in log-RMSE against 0.181 for a power law refitted inside the same folds on the same rows and features, a margin of 29%, and against 0.199 for the published IPB98(y,2) law, a margin of 36%.

The two comparators are not interchangeable and both are quoted throughout. IPB98(y,2) is a historical reference: it was derived in 1998 from version 2.8 of this database's predecessor (DB2.8) rather than from the DB5.2.3 revision analysed here [11]. That predecessor overlaps the present database substantially in devices and operating regimes, so for most folds below the machine being held out contributed to the data IPB98(y,2)'s exponents were fitted on. It is therefore not a fold-specific comparator that can be described as having been fitted without the held-out rows, and no claim in this paper rests on treating it as one. The power law refitted inside each fold is the like-for-like baseline, and the reversal below is stated against it. Every number in this paper is scored on the nine engineering features; the ten-column matrix of Sec. 3, which adds IPB98(y,2)'s own prediction to them, is reported there as a control and nowhere else.

Because the result is a negative one about a common practice, it is worth saying early what a reader can check rather than take on trust. Every number and figure here regenerates from the published OSF deposit, which is verified against a pinned SHA-256 on load rather than redistributed, at the single commit named in Data availability; and the test suite binds each figure quoted in the prose to the field of the generated file it came from, in both directions, so a rerun that moved a value would fail rather than pass silently. Nothing below asks to be believed.

Related work on this database. Four recent studies apply machine learning to the same ITPA data and are the closest prior work. Nam and Seo [18] take cross-machine transfer as their subject: they align features to correct the covariate shift between devices and use deep ensembles for uncertainty, validating on conditions the model has not seen, including JET-ILW and ITER. Gao *et al.* [19] apply variational and sampling-based Bayesian neural networks to STD5 together with individual JET, DIII-D and ASDEX Upgrade data, reporting improved accuracy alongside posterior uncertainty. Xiong *et al.* [20] tune random forests and two neural architectures by Bayesian optimisation on STD5 and report R^2 up to 0.990, with a feature-importance analysis separating spherical from conventional tokamaks. Wang *et al.* [17] anchor a Kolmogorov–Arnold network on a frozen power law and learn only the residual, on this same DB5.2.3 revision. Their unanchored network interpolates accurately and, in their words, behaves unpredictably outside the training distribution, while anchoring it to a power-law trend substantially improves its stability across held-out cohorts defined by parameter ranges. They also try a Mahalanobis-distance gate at prediction time and report that it helps in some shifted regions, is not universally beneficial, and cannot compensate for missing device coverage.

That last work deserves to be stated precisely, because the shape of the result here is not new with this paper. A flexible model that wins the interpolation comparison and degrades badly out of distribution, repaired by supplying a power-law trend it cannot invent for itself, is what Wang *et al.* report, on this database, before this paper. What is new here is the unit that is held out. Their cohorts are defined by thresholds on parameter ranges, which is a shift constructed inside the population of machines the model has already seen; the deployment claim a scaling law makes is about a machine that is not in that population at all, and they say explicitly that device coverage is what their gate cannot substitute for. Nam and Seo [18] likewise validate on unseen conditions and report that cross-machine transfer is harder.

So the contribution is not the first report of a reversal. It is that the reversal is measured with the device as the held-out unit, on every device the database can score; that the ranking inverts exactly rather than narrowing; that the calibrated interval is measured at the same time and fails without any signal in its width; and that flexibility and long-range saturation are separated, which says which property to avoid rather than which model. Accuracy degrading out of distribution is a matter of degree and a margin can absorb it. A reversed ranking over the devices a law is deployed on cannot be absorbed by a margin, and neither can an interval that stops covering at unchanged width.

The difference is not one of degree. Selecting on the cross-validated score is what the convention prescribes, and applied to the three primary models here it picks the one that extrapolates worst. A practitioner comparing that set is therefore not left with an open question about extrapolation. They are led to the worst available answer, and the more carefully they follow the convention the more confidently they are led there. The scope of that is the candidate set rather than the score: Sec. 12 fits a model with comparable flexibility and a different asymptote, and it is at once the best cross-validated model in this work and one that extrapolates, so widening the set repairs the rule rather than replacing it. Degradation is expected and can be budgeted for; a selection rule anti-correlated with the goal it is used for, over the models actually being chosen between, cannot.

None of these works is the target of a criticism here; what any of them reports is what its split measures. Where a score on this database is computed without holding out a whole device, it describes discharges from machines the model has already seen. No better predictor is proposed here. What is asked is whether the validation that selects one answers the question a scaling law exists for, what the honest answer costs, which property of a model decides it, and whether physics rather than flexibility supplies the repair.

That measurement does not directly address the deployment problem of interest, for a reason about the data rather than about the models. Cross-validation on this database holds out *discharges*, and every machine in a held-out fold also appears in the training fold, so a held-out row has hundreds of near neighbours recorded on the same device at nearby settings. What the score quantifies is how much of JET is predictable from the rest of JET. That a validation split should match the generalisation being claimed, and that grouped observations flatter a model when it does not, is well understood in fields whose data arrives in clusters [21]; it is not what a commonly reported comparison on this database measures. Inside this field, Abbate *et al.* [10] make the same argument for transport models rather than scaling laws: that a model has to be scored on a large database against a baseline chosen for the use it is put to, and that on that footing two integrated codes show no significant advantage over a two-parameter empirical baseline for temperature profiles. Materials informatics reached the same design from the same motive and is the closer methodological precedent: leave-one-cluster-out cross-validation was adopted there specifically to measure extrapolation to unseen chemistries rather than interpolation within seen ones [7], and reporting how far a target sits from the training distribution is the applicability-domain question that field has treated as standard for two decades [8]. Neither the split nor the distance diagnostic used below is new outside fusion; what is new is what they measure on this database. The quantity a scaling law is built to supply is a prediction for a machine that does not yet exist.

This paper measures what the deployment-matched split costs. What is held out is escalated in three steps, a discharge, then a database label, then a physical device, and the same models are scored under each. A fourth comparison holds out an entire size range; its held-out side is 96% one physical machine, so it is reported as a stress test across one large jump rather than as a fourth rung of the same ladder, and no claim here rests on it alone. Under the second step the ranking of the three primary fitted models is the exact reverse of their ranking under grouped cross-validation, and the best cross-validated model *of those three* loses to a log-linear power law on all 13 held-out database labels and all 11 physical devices. That qualifier is load-bearing: the best cross-validated model in this work is the Gaussian process of Sec. 12, at 0.112, and it does not reverse. Three measurements say why, and they cover extrapolation distance, the training-target bound of a random forest, and what polynomial flexibility costs in the tail. At the size cut, which reproduces ITER's extrapolation factor inside the database, conformal intervals calibrated to 90% cover 3% of rows without becoming any wider. Two repairs then supply the long-range structure the diagnosis implicates: the Connor–Taylor dimensional constraints imposed on the exponents, and a Gaussian-process experiment that separates flexibility from long-range saturation and corrects the diagnosis the earlier sections leave open. The result is replicated on rows the standard analysis set does not contain, in a second confinement regime, and, in Secs. S4 and S5 of the Supplementary Material, outside fusion. Section 3 begins with a property of the design matrix itself: it is rank deficient by two, and one of the two dependencies is that a published scaling law added as a model feature contributes nothing a log-linear fit can use.

2 Data and methods

Data. The ITPA Global H-mode Confinement Database, standard analysis set STD5 v5.2.3 [11], obtained from the Open Science Framework (OSF). The delivered file contains 6228 rows from 4471 discharges across 18 database device/wall labels, which correspond to 16 physical devices once the two JET entries and the two ASDEX Upgrade entries are each collapsed. The cleaning rule below requires finiteness and positivity in every used column; on this file it removes no rows, so all 6228 delivered rows are analysed. The target is the thermal confinement time `TAUTH`. No synthetic data appears in any result. The analysed file is pinned by SHA-256 and verified on load, so every number below refers to one specific set of bytes rather than to whatever the host currently serves.

Features and models. Nine log engineering features: plasma current, toroidal field, line-averaged density, loss power, major radius, elongation, inverse aspect ratio, effective mass and minor radius. The analytic IPB98 prior is *excluded* as a feature throughout the extrapolation arms. Its exponents were fitted on a predecessor of this database that overlaps it in devices and regimes, so for most folds it carries information about the held-out label; retaining it would import that overlap into every fold. The model set is a ridge regression on the log features (i.e. a fitted power law), a random forest (300 trees), a histogram gradient booster, polynomial ridge controls of degree 2 and 3, and a mean baseline. The forest is bagged trees in the sense of Ref. [22] with `max_features` = 1.0, so every split considers all nine features rather than a random subset; that is the scikit-learn default for regression and is stated because it is a choice, and because Sec. 4.2 shows a nested search moves off it in 13 of 19 folds and improves the held-out score when it does. IPB98(y,2) evaluated analytically is the historical reference of Sec. 1, marked as such wherever it is scored, and its scores are not part of the ranking claim.

What is and is not tuned. Every model is fitted and scored in $\log\tau$: the target is $\log\tau$ and the reported error is the root-mean-square residual in the same variable. Exponentiation to τ enters only where a quantity is quoted in seconds, such as the mean absolute error and the device forecasts of Sec. 11. None of the models in Table 1 is hyperparameter-tuned, and "library defaults" is not a specification, so the consequential settings of every model, and of the polynomial and Gaussian-process rungs, are written out in full in Sec. S7 of the Supplementary Material rather than left to a default a later release may change.

Every estimator carries the seed 42, and the forest is fitted in parallel but predicts single-threaded so that reruns are bit-identical. Standardisation sits inside the pipeline, so it is refitted on the training rows of every fold and never sees a held-out row. No within-model hyperparameter search is performed on any reported cross-validation score, so nested tuning is not required for those numbers, and Sec. 4.2 reports what a nested search does when one is run anyway. Comparing several prespecified model families is still a selection of a kind, and two later choices go further: the constraint rung of Sec. 8 and the damping of Sec. 6 were both picked after seeing held-out scores, and both are flagged where they appear. The Gaussian processes of Sec. 12 do estimate kernel hyperparameters, by marginal likelihood on training rows alone.

Splits. (i) Grouped cross-validation by discharge: `GroupKFold` with five folds, grouped on the discharge identifier, so every slice from one discharge falls in the same fold. It is constructed with `shuffle=False`, its default, so the partition is deterministic and no seed enters it; Sec. 4.2 reports the same comparison over shuffled partitions. (ii) Holding out a whole unit, in two versions that are named separately throughout because the 18 labels are not 18 physical devices. *Leave-one-label-out* (LOLO in the tables) holds out each of the 13 database device/wall labels with at least 30 rows in turn, training on every other row in the database. *Leave-one-device-out* holds out each of the 11 physical devices those labels reduce to once JET and JET-ILW are merged, as are the two ASDEX Upgrade entries. The distinction matters here more than it usually would: holding out JET-ILW while JET stays in training does not hold out an unseen device, and this paper's own argument is that the held-out unit has to match the deployment being claimed. Leave-one-label-out is the default reported column because it is the finer partition and the one a practitioner would reach for first; every claim that turns on a device being genuinely unseen is stated against leave-one-device-out and labelled as such. The 30-row threshold governs only which labels are scored, not which are trained on, so the five labels below it (COMPASS, TCV, START, TDEV and TFTR, 43 rows between them) stay in every training fold; each fold therefore trains on 17 labels. It was fixed before any held-out score was computed, as the smallest round number giving a per-label log-RMSE over enough rows to be worth reporting, and it is not a tuned quantity: Sec. 4.2 sweeps it over 10, 20, 30 and 50 rows and reports what the win count does, and also reports what happens when the five sub-threshold labels are dropped from training as well. (iii) A size-ordered cut: labels are ordered by median major radius, the model is trained below a cut and scored on every label above it, and the reported cut is the one whose size ratio is closest in log to the ratio between this database and ITER. That rung has a ratio of 1.823, trains on 14 database labels and scores 2730 held-out rows, and is called the ITER-size-matched cut throughout. The match is to ITER's major-radius ratio and to nothing else: ITER is a burning plasma dominated by alpha self-heating at a gain no row here approaches, so clearing this bar is necessary and not sufficient.

The score is the root-mean-square error in $\log\tau$, $\mathrm{RMSE}_{\log\tau} = \sqrt{\overline{(\log\hat{\tau} - \log\tau)^2}}$, written log-RMSE here and in the figures. It is the plain log residual, not the $\log(1+y)$ quantity usually called RMSLE; confinement times are strictly positive, so no offset is needed, and the name is avoided here to keep the two distinct. Intervals are percentile bootstraps resampling whole *units*, since the claim concerns behaviour on an unseen unit: 2000 resamples drawn with replacement over the 13 labels, seeded at 20240617, reported as the 2.5th and 97.5th percentiles. Paired intervals resample the per-label difference rather than the two error levels, which removes the shared unit difficulty.

What the inferential statistics here do and do not claim. Stated once, and referred back to rather than repeated. Every fold in a leave-one-unit-out arm trains on all the other units, so any two folds share almost all of their rows. The units they score are therefore not independent experiments, and nothing below treats them as such. Three consequences follow and hold throughout. The bootstrap intervals are summaries of spread over the held-out units, not confidence intervals under independent replication. The permutation and sign-test p-values, including the exact sign test under what is called the independent-unit-sign null in Sec. 4.2, are compact summaries of how lopsided a win count is, not evidence from n replicates. And the conformal coverages are empirical row-level rates rather than guaranteed ones, for the separate reason given in Sec. 7. What carries the claims in this paper is the descriptive win counts and the size of the gaps, which need none of that machinery; the statistics describe their spread.

Preprocessing. Rows are dropped only for non-finiteness or non-positivity in a used column, which on this file removes none of the 6228 delivered rows; no outlier rule, no winsorising and no reweighting is applied. Features are natural logs of the raw engineering columns, taken after cleaning. Minor radius is reconstructed as $a = \epsilon R$ at this stage, which is the origin of one of the two exact collinearities of Sec. 3. Wall variants (JET-C and JET-ILW, and the two ASDEX Upgrade entries) are separate labels in the database and are kept separate except where the text says devices are collapsed to 11.

Software. Python 3.12.14, scikit-learn 1.7.2, NumPy 2.2.6, SciPy 1.18.1 and pandas 2.3.3. The version of every library is pinned in the repository's `constraints.txt`. Any setting left at its library default therefore resolves to those versions' default; the consequential settings are written out above rather than referenced, because a default may differ in a later release.

3 The design matrix, and what a refit can identify

Three column counts appear in this paper and they are not the same set. The models see nine log engineering features. Adding the analytic IPB98 prediction to those nine gives the ten-column matrix examined in this paragraph. The refit of Eq. (1) in Sec. 3.1 drops both redundant columns and uses nine, eight independent variables plus an intercept, which is why its condition number is a healthy 10.7.

Standardised, the ten-feature matrix has rank 8, with the last two singular values at 4.9×10^{-14} and 2.4×10^{-14}. Projecting candidate vectors onto the null space confirms two exact dependencies at relative residuals of order 10^{-16}: minor radius is derived as $a = \epsilon R$ during cleaning, and the IPB98 prediction is a fixed log-linear combination of the other eight features. The second is the substantive one. Adding a published physics scaling as a model feature feels like adding knowledge; in log space it provably is not, and it contributes exactly nothing to a log-linear fit. The deficiency raises no error, because `scipy.linalg.lstsq` inverts through the SVD pseudoinverse and returns the minimum-norm member of a two-parameter family of coefficient vectors that fit identically well.

Neither the redundant column nor the estimator is load-bearing. Dropping `log_a_m` and refitting on eight independent columns leaves the log-linear scores unchanged to four decimals under all three splits, moves the tree ensembles by at most 0.035, and changes no ordering anywhere. Ordinary least squares on those same eight columns, carrying no penalty at all, reproduces the reported ridge to six decimals under cross-validation and to 0.0012 at the size cut, so the ridge here is a numerically stable spelling of the least-squares power law rather than a regularised alternative to it. The models keep the redundant nine-column representation because minor radius is a parameter a practitioner would naturally supply, and promoting the cleaner eight-column variant on the strength of its scores would be exactly the selection Sec. 8 flags in its own result.

3.1 Refitted exponents and a weakly identified direction

Refitting Eq. (1) on all STD5 rows (design matrix 6228×9, condition number 10.7) gives I_p 1.08 against the published 0.93 and R 1.58 against 1.97, while P and B_t return almost exactly. The disagreement is expected: IPB98 was fitted to a selected ITER-relevant subset, whereas this fit applies no selection criteria and includes the spherical tokamaks, and a refit on a different population is a different regression. It is not a solver artefact either; independent implementations by the normal equations, by QR and by the SVD pseudoinverse agree to 7.6×10^{-13}.

Where the two differ matters more than by how much. Decomposing the difference between the exponent vectors along the singular directions of the design matrix, 77% of it lies in the single weakest direction, which carries 0.3% of the matrix's variance, while the three strongest directions carry 82% of the variance and account for 0.75% of the disagreement. That weak direction is plasma current traded against machine size, and it is weak for a structural reason: tokamaks are not designed to vary the two independently, so the existing multi-machine design leaves the direction weakly identified. More shots at the operating points these machines already run would not separate it; shots deliberately placed to break the correlation could.

Why the size dependence in this database is weaker than IPB98(y,2)'s is itself an active question, and Hall *et al.* [23] attack it directly on DB5.2.3, treating the multicollinearity among the predictors explicitly and searching for the smallest subset of points that most reduces the fitted major-radius exponent. A continuation [24] argues, in a conference presentation, that the weak size dependence is attributable predominantly to a subset of discharges localised in dimensionless space, distinguished by normalised pressure, collisionality, normalised gyroradius and safety factor rather than by any single device or by metallic walls, and reports that excluding it recovers a major-radius dependence broadly consistent with IPB98(y,2). That is a different question from the one asked here. They ask where the changing size dependence comes from; this paper takes the fitted exponents as given and asks how the choice of validation split changes which model is preferred. The two bear on each other: if the size direction is partly an artefact of a localised subset, then the size-ordered cut of Sec. 5 measures transfer across a boundary those authors would draw differently. That is a caveat on how much weight that one axis can carry, and not on the ranking reversal, which is measured per unit rather than per size.

4 The ranking inversion

Table 1 is the central result. Among the three primary models, a power law, a random forest and a gradient booster, the order under one split is the exact reverse of the order under the other. The random forest is the best of the three primary models under cross-validation and the worst of the three on a label it has not seen. It is not the best model in the paper: the Gaussian process of Sec. 12 scores 0.112. Nor does that model reverse. On a held-out label it scores 0.218 against the power law's 0.214, which is a tie and not a win, but it is nothing like the forest's 0.465, and across the ITER-size-matched cut it scores 0.191 against the power law's 0.278. So cross-validated selection over the full candidate set of this work does not choose the forest at all. The inversion is a statement about the three primary models, and Sec. 13 takes up what that costs the general claim. Against the fold-fitted power law the forest is 29% better in the CV column and 117% worse in the LOLO column. The analytic IPB98(y,2) row is the historical reference of Sec. 1 and carries no rank. The two polynomial rungs are controls, scored in the same table but outside the ranking claim, and including them breaks the inversion rather than extending it: the log-cubic rung is third of the five fitted models under cross-validation and last of them on a held-out label.

Because the labels differ enormously in difficulty, the marginal bootstrap intervals overlap and are the wrong test; resampling the *paired* per-label difference removes that shared difficulty. The random forest is worse than the log-linear power law on **13 of 13** database labels, mean gap +0.251, 95% interval $[+0.157, +0.342]$. The gradient booster is worse on 12 of 13. Figure 1 shows the reversal together with the three extrapolation diagnostics of Sec. 4.1. Section 4.2 repeats this over the 11 physical devices, where the labels are no longer two eras of one machine counted twice, and the gap widens to +0.315 with an interval of $[+0.221, +0.408]$.

4.1 Three diagnostics of the failure

Distance. Ranking each model's per-label error against the Mahalanobis distance of that label's mean log-feature vector from the training mean gives $\rho = +0.85$ for the random forest and +0.54 for the gradient booster, against $\rho = -0.06$ for the log-linear power

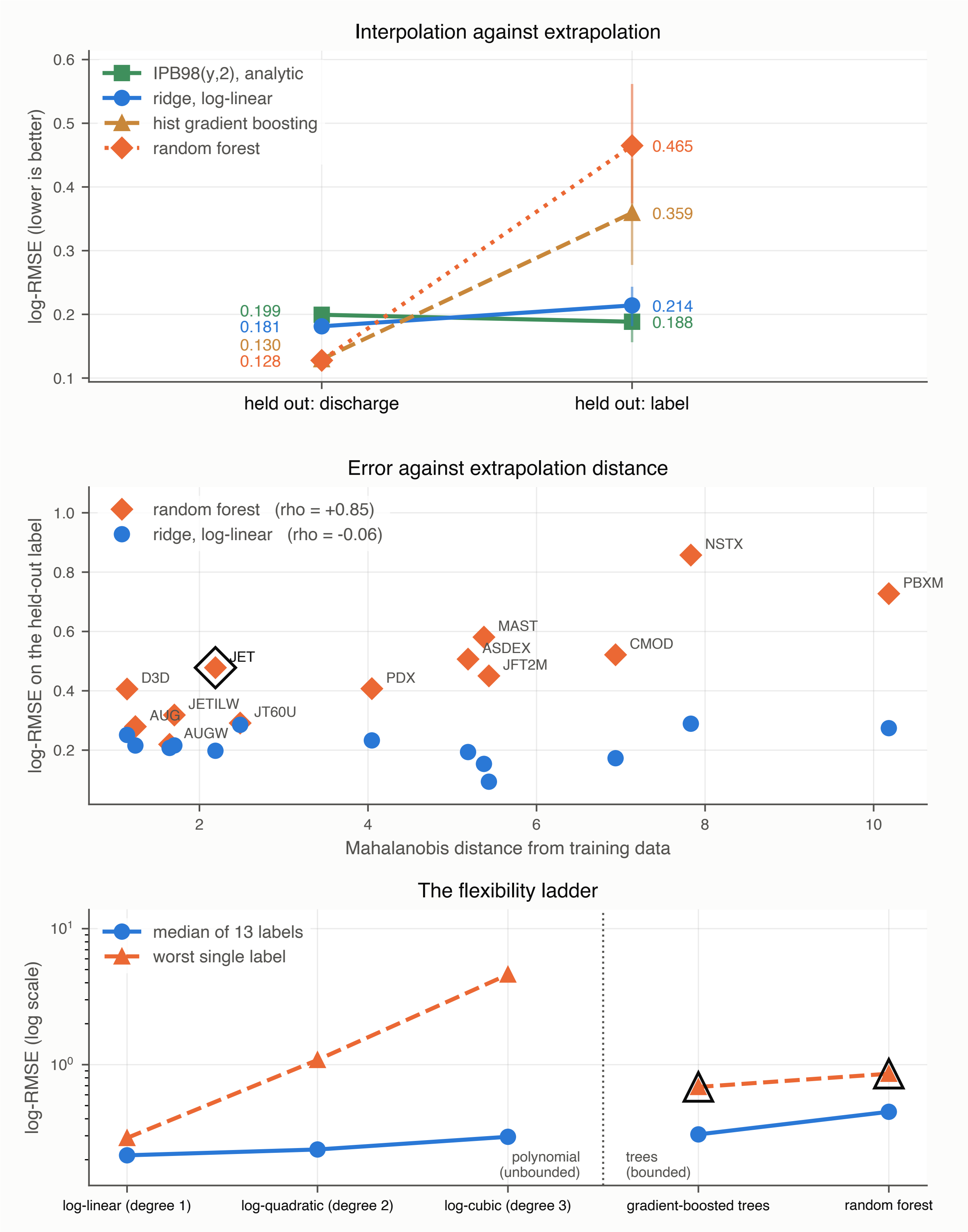

Figure 1: Interpolation against extrapolation. Top: each model under both splits, with bars giving the 95% interval over the 13 held-out labels; the lines crossing is the result. Middle: per-label error against Mahalanobis distance from the training data, rising strongly for the random forest, more weakly for the booster, and flat for the power law. The circled label is the second failure mode, where the true confinement times run above anything the random forest can output. Bottom: the flexibility ladder, where what grows with flexibility is the tail rather than the centre; the circled points are the two tree ensembles, both observed here to stay inside the training-target range, though only the random forest is provably bounded to it.

Table 1: The headline comparison under the conventional reporting choices: the same models and the same nine engineering features, scored one way per split (LOLO is leave-one-label-out over the 13 eligible labels). The two columns differ in more than the held-out unit, since they also differ in row population and in how per-row scores are aggregated, and the 13 labels are not 13 independent devices. Sec. 4.2 matches the population and the aggregation and repeats the comparison over physical devices; the inversion survives all of it. The rank columns cover only the six models above the rule, all of which are refitted inside every fold. IPB98(y,2) is set below the rule and carries no rank: it is not refitted within folds and is not part of the ranking claim.

Model	CV	LOLO	Ratio	CV rank	LOLO rank
random forest	0.128	0.465	3.6×	1	4
hist gradient boosting	0.130	0.359	2.8×	2	3
ridge, log-cubic	0.148	0.849	5.7×	3	5
ridge, log-quadratic	0.158	0.300	1.9×	4	2
ridge, log-linear	0.181	0.214	1.2×	5	1
mean baseline	0.869	0.994	1.1×	6	6
Historical reference, not fold-fitted					
IPB98(y,2), analytic	0.199	0.188	1.0×	n/a	n/a

law. The random forest's error rises strongly with extrapolation distance and the gradient booster's trends upward more weakly. The power law's does not: its error is uncorrelated with how far the label sits from anything it saw.

Thirteen points is few enough that each of those numbers needs an uncertainty. Permuting the distances 20000 times gives $p = 0.0007$ for the forest, $p = 0.060$ for the booster and $p = 0.85$ for the power law. That permutation treats the 13 database labels as exchangeable, which they are not exactly: two physical devices contribute two wall-era labels each, so these p-values are read as Sec. 2 says: summaries of how lopsided a count is. Recomputed over the 11 physical devices, which removes the duplicated wall-era labels, the forest's correlation weakens slightly to $+0.77$ ($p = 0.005$) and the power law's is unchanged at -0.06 ($p = 0.85$), so the contrast the section rests on survives the collapse. That permutation p-value is read the same way. The booster's falls from $+0.54$ to $+0.38$ ($p = 0.25$). Neither figure clears a conventional threshold, and the booster's correlation is read throughout as suggestive rather than established. Dropping each label in turn moves the forest's ρ only within $[+0.81, +0.88]$, so no single extreme device carries it. One control changes how the rest should be read: the *mean baseline*, which fits nothing, also correlates with distance, at $+0.56$. Distance predicts how hard a unit is in general. What is unusual is not that the trees degrade with it but that the power law does not.

The same distance in dimensionless coordinates. Organising an extrapolation audit by engineering variables invites one objection, and it is the objection Hall *et al.* [24] have raised in a conference presentation: the weak size dependence in this database may come from a subset localised in *dimensionless* space rather than from size. Sec. S12 of the Supplementary Material repeats the distance construction above in (ρ^*, β, ν^*), built from the same group definitions the Connor–Taylor constraints of Sec. 8 are derived from and a temperature recovered from the stored energy of Sec. 4.2. The diagnosis translates: the two distance rankings agree at $\rho = +0.78$, and on the rows a placement is possible for the forest's error tracks dimensionless distance at $+0.77$ against $+0.79$ for engineering distance, with the power law flat against both. The ITER-size-matched cut does not translate, and that half goes to Hall: across it $\log \rho^*$ and $\log \nu^*$ separate as well as size does, so that cut is a joint displacement in size, normalised gyroradius and collisionality rather than a size boundary. Sec. 5 says so, and the leave-one-device-out comparison this paper rests on carries no such confound.

Mahalanobis distance between mean log-feature vectors is one measure of how far a unit sits from the training set, and it is the only one used here. It compares first moments through a single pooled covariance, so it is insensitive to differences in the shape or spread of a unit's operating envelope at a fixed mean, and the covariance it uses is singular by Sec. 3 and so is inverted through the pseudo-inverse. Every statement below about "distance" is a statement about that one diagnostic, not about distributional difference in general.

The per-label scores behind that correlation are in Sec. S8 of the Supplementary Material. On the five most distant labels (PBX-M, NSTX, C-Mod, JFT-2M and MAST, all small, two of them spherical) the forest's error is 2.7 to 4.8× the power law's; on the four nearest it is within 1.7×. JFT-2M and MAST sit within 1% of each other in distance, so the boundary is drawn to be insensitive to that.

A hard bound, and the one fold it binds on. A random forest predicts by averaging leaf values, and every leaf value is itself a mean of training targets, so every prediction it can make lies in $[\min y_{\text{train}}, \max y_{\text{train}}]$ whatever the features say. With JET held out, 48% of its rows exceed the maximum confinement time anywhere in the training fold, and its largest held-out target is 3.7× above anything any tree in the forest can output. More rows would close this only if they carried larger targets: within the same target range no amount of additional data, and no tuning, moves a ceiling that is a property of the functional form.

That is one fold of thirteen, and where the bound is active is reported here rather than left to be found, because it decides which reading of Sec. 12 this diagnostic supports. JET is the only held-out label whose targets reach the ceiling at all. On the other twelve the largest held-out target sits below it, by 1.31 in logs on JET-ILW and by 3.27, a factor of 26, on MAST; `results/boundedness.json` carries the margin for every fold. The size cut of Sec. 5 is the second place it binds, and there **34%** of held-out rows lie above the ceiling.

Set against the per-label gaps, that pattern is evidence, and it points away from the bound rather than toward it. The four labels the forest loses by the widest margins are NSTX (+0.568), PBX-M (+0.453), MAST (+0.427) and JFT-2M (+0.357), and on each of them its ceiling stands about three natural logs above anything the held-out machine reaches. JET, the one fold where the ceiling is in the way, is seventh of the thirteen by margin, at +0.280. The forest therefore fails hardest where the bound is furthest from binding, and a constraint that is not active cannot be what an error is made of. What is left has to operate in the interior of the training range: predictions that stop tracking the features once the input leaves the support the model was fitted on, whether or not a ceiling is anywhere nearby. Section 12 builds the model that separates the two readings and finds the second one, and this is the first evidence in the paper that points there.

A gradient booster carries no such guarantee. It sums tree outputs onto an initial estimate rather than averaging targets, and nothing in that construction confines the sum to the training range, so the argument above does not transfer to it. What can be reported is that it never uses the freedom: over the 13 leave-one-out folds and the size cut of Sec. 5, its largest prediction stays below $\max y_{\text{train}}$ on every one, coming closest with JET held out at 0.066 in logs beneath the ceiling and sitting 0.567 beneath it across the size cut. Its boundedness on these rows is measured rather than guaranteed, and every claim below that needs the guarantee is made about the forest.

Flexibility costs the tail, not the centre. A polynomial ladder in the log features (degrees 1–3) is flexible but still extrapolates. Degree 2 is *not* meaningfully worse than degree 1 on a typical label: median 0.238 against 0.216, better on 8 of 13, paired interval $[-0.013, +0.237]$ containing zero. What flexibility costs is the tail. Degree 1's worst label is 0.289; degree 2's is 1.083 and degree 3's is 4.601, both on C-Mod. A bounded range is simultaneously why neither ensemble extrapolates and why neither blows up: their worst cases, 0.686 and 0.857, sit far below degree 3's.

Read together, these three diagnostics invite a summary this paper does not endorse. Every flexible model scored so far is also either bounded, like the two ensembles, or divergent, like the cubic, so nothing above can tell apart "flexible models fail out of distribution" from "models that stop trending outside their data fail out of distribution". The two make identical predictions about everything in Table 1. Section 12 builds the model that separates them and supports the second interpretation, so the reader should carry the flexibility reading as provisional from here rather than as the paper's conclusion.

4.2 Sensitivity to population, aggregation, and device definition

Table 1 attributes a difference to the split, so anything else that differs between its two columns is a confound. Three things do. Grouped CV scores all 6228 rows while leave-one-out scores only the 13 labels large enough to hold out. Grouped CV pools out-of-fold predictions into one log-RMSE over every row, where JET and AUG dominate, while leave-one-out averages per-label scores, where every label counts once. And JET and JET-ILW are one physical tokamak either side of a wall change, as are AUG and AUGW, so holding out JET-ILW while training on JET does not hold out an unseen device.

The labels too small to score. A fourth choice is worth stating because it looks like a confound and is not. Five labels hold fewer than 30 rows (COMPASS, TCV, START, TDEV and TFTR, 43 rows between them, under 1% of the database) and none can carry a held-out score of its own. They are nonetheless in every training fold, since the threshold governs scoring rather than training. They are also among the most unusual points in the feature space, so whether a flexible model can exploit those few rows is a fair question to ask of the ranking.

Dropping them from training as well answers it. Every fitted model gets slightly worse without them, by 0.012 for the power law, 0.027 for the gradient booster and 0.007 for the random forest, and the ordering over the 13 held-out labels is unchanged: power law 0.214 to 0.226, gradient booster 0.359 to 0.387, random forest 0.465 to 0.472. The 43 rows help the tree ensembles slightly, as one would expect of rows in a sparse region, and nowhere near enough to recover the inversion. The analytic reference is unaffected, since it fits nothing.

Table 2 varies all three. The forest leads the power law under every cross-validation cell and trails it under every leave-one-out cell, so the inversion is a property of the split rather than of the population or the aggregation. Collapsing the variants makes the forest's failure worse rather than better: over 11 physical devices its mean gap over the power law grows from +0.251 to +0.315, with a paired 95% interval of $[+0.221, +0.408]$ against $[+0.157, +0.342]$ over the labels, and it loses on 11 of 11. The published law's score is almost unmoved by any of it, 0.188 to 0.184 per unit.

Table 2: The random forest and the power law under both row populations, both aggregations, and both definitions of a machine. The inversion is the pattern of the random forest leading under cross-validation and trailing the power law under leave-one-out; it holds in all eight combinations.

	pooled over rows		each unit equally	
Split	forest	ridge	forest	ridge
CV, all 18 labels	0.128	0.181	0.149	0.238
CV, the 13 scored labels	0.123	0.177	0.124	0.187
leave-one-label-out (13)	0.410	0.214	0.465	0.214
leave-one-device-out (11)	0.551	0.200	0.527	0.211

A newer published law. IPB98(y,2) is the ITER reference but not the most recent scaling fitted to this database family. ITPA20 and ITPA20-IL [11] were derived from DB5.2.3 and weaken the size dependence considerably, so they are the obvious second reference. Each was fitted to a different selection out of DB5.2.3, and each is specified on the separatrix elongation at the geometric axis, κ, where the delivered file carries one elongation column, `KAPPAA`: the areal elongation $\kappa_a = S/(\pi a^2)$ computed from the plasma cross-sectional area, which is the definition IPB98(y,2) was fitted to and is the column loaded here.

That mismatch does not have to be modelled, because it can be measured. STD5 is an extract; the full DB5.2.3 revision this paper already pins carries the boundary elongation `KAPPA` on every analysed row, joined on machine, shot and time by the procedure the stored-energy control below specifies and checks. So ITPA20 and ITPA20-IL are scored below on the elongation they were fitted to, and IPB98(y,2) on the areal elongation it was fitted to, which is what makes the three comparable at all.

What the substitution costs is worth stating, and it is worth stating that a shape model is the wrong way to find out. On a parametrised boundary κ/κ_a depends on the triangularity alone, which confines the ratio to $[1.000, 1.110]$. The measured ratio has a median of **1.083** and a maximum of **1.341**, with 16% of rows above that ceiling and 7% *below* unity, which no boundary of that family produces at all. Only **2.0%** are below it by more than 5×10^{-4}; the rest is rounding on machines that ran circular.

The rows below unity are worth following, because the obvious reading of them is that the delivered column is wrong and the obvious reading is not right here. A boundary elongation below the areal elongation is impossible for a *convex* boundary, and of the 2.0% that clear rounding, all but PBX-M's 59 rows sit within 3.5% of unity on near-circular machines, which the two columns' precision covers. PBX-M does not: its ratio runs from 0.67 to 0.80, at a median of **0.739**. PBX-M ran indented, bean-shaped plasmas, its boundary is not convex, and the convexity argument does not apply to it. DB5.2.3 records that directly: of the analysed rows, PBX-M is the only label carrying an indentation above 0.04, at a median of **0.154**, and the 60 rows with any indentation at all have a median κ/κ_a of 0.740. The column is not defective on those rows; the parametrised boundary is simply the wrong shape for them, which is the same point the ratio makes everywhere else and is here checkable against a third column.

Scoring on the delivered column rather than the measured one moves ITPA20 by up to 0.018 and ITPA20-IL by up to 0.022, twenty to thirty times what the model would predict and enough to reorder ITPA20 against IPB98(y,2) per label. Where a column exists, a model of it is not evidence about it.

Evaluated analytically on the rows analysed here, ITPA20 scores 0.188 over all rows, 0.191 per label and **0.177** across the ITER-size-matched cut; ITPA20-IL scores 0.237, 0.269 and **0.165**. Against IPB98(y,2)'s 0.199, 0.188 and 0.194, ITPA20 is below over all rows and at the size cut and level with it per label; ITPA20-IL is above it at the first two and below it at the cut, where it is the lowest of the three. Both are below the power law refitted inside these folds at that cut, which scores 0.278. Against the random forest's 0.938 and the gradient booster's 1.072 there, ITPA20 is better by factors of 5.3 and 6.1.

That row is not blind and is not offered as one. ITPA20 and ITPA20-IL were fitted to selections out of DB5.2.3, so at the size cut they had seen the large machines the fitted models are being asked to reach, and their margin over IPB98(y,2), which was fitted to the DB2.8 predecessor, is at least partly that. What the comparison does carry is that no published power law in this family lands anywhere near where the tree ensembles land, and that the two fitted to this revision are the two that beat a refit of the same functional form on these rows.

Do the features name the machine? One objection is more serious than the four below it, because it explains everything in Table 1 without any appeal to long-range behaviour. Four of the nine features barely move once a device is fixed. Major radius carries **0.1%** of its log variance within devices, minor radius 1.1%, inverse aspect ratio 4.9% and elongation 6.1%. A random forest classifier recovers the device from the nine log features at **99.9%** under the same discharge-grouped split used throughout, against a 42.3% majority baseline. The feature vector is very nearly a device name. Under grouped cross-validation a tree can therefore split on those coordinates and fit a per-device offset, which is a lookup table: worth a great deal in sample, and worth nothing on a machine that is not in it.

Deleting those features is the obvious control and it cannot settle the question, which is worth showing rather than asserting. Refitting on the four features that do move within a device, plasma current, toroidal field, density and loss power, the reversal disappears: the forest's interpolation margin *grows* to 50.3% and it is worse than the power law on 6 of 13 labels and 5 of 11 devices, at mean gaps of −0.033 and −0.018. That looks decisive and is not, because the deleted columns carry the size dependence, and the size dependence is the thing the power law extrapolates through. The control removes the suspected leak and the physics under test together, and the power law's held-out score more than doubles, from 0.214 to 0.477. It tells us that the reversal needs features carrying size. It cannot tell us whether it needs them because they carry size or because they name the machine.

The dimensionless coordinates separate exactly those two things. In $(\rho^*, \beta, \nu^*, q, \epsilon, \kappa, M)$ the size, field and density dependence is retained while no coordinate is a device label: ρ^* moves 26.6% within a device, β 35.4%, ν^* 66.6% and q 70.4%, against major radius's 0.1%. Regressing $B\tau_E$ on those seven groups, with the same models and the same splits, the reversal survives. The forest wins cross-validation by **28.9%**, against 28.7% on the engineering features, and loses on **11 of 13** labels and **9 of 11** devices, at mean gaps of +0.074 and +0.130. The margins are smaller than the engineering-space ones and the direction and the counts are the same.

So the reversal is not an artefact of the features naming the machine. It is also not established at the strength the engineering-space numbers suggest: a part of that +0.315 is device identity, and +0.130 is what survives in coordinates where identity is unavailable. Both numbers are reported, and the dimensionless one is the conservative reading. Two limits attach to it. The temperature it needs comes from the stored energy of the control above, so it is scored on 6214 rows rather than 6228; and ϵ and κ still move only 5% and 6% within a device, so the coordinates are cleaner than the engineering ones rather than clean. The specification is in `analysis_dimensionless.py`.

Four choices that do not carry it. Four more arms vary choices the comparison had to make and that a reader could reasonably suspect of carrying it. Sec. S11 of the Supplementary Material gives each in full; the outcomes are that none of them does. Sweeping the 30-row eligibility threshold over 10, 20, 30 and 50 rows changes only which labels are scored and never which are trained on, and the power law wins every scored label at 20, 30 and 50 and 14 of 15 at 10. Repeating the cross-validation over ten shuffled fold assignments puts the forest's interpolation margin between 27.0% and 28.5%, with the deterministic partition's 29.5% just above that range, so the margin it later gives up is not a property of one partition. A nested hyperparameter search moves the forest's settings in 13 of 19 folds and improves its held-out label score from 0.465 to 0.403, or 0.376 with the inner folds grouped by machine to match deployment, against the power law's 0.214; at the size cut it makes the forest worse. And dropping the spherical tokamaks, which IPB98(y,2)'s own selection excluded, leaves the forest worse than the power law on all 11 remaining labels.

The loss power is the target's denominator. The thermal confinement time is the thermal stored energy over the thermal loss power, so $\log \tau_{\mathrm{th}} = \log W_{\mathrm{th}} - \log P_{\mathrm{LTH}}$ up to the deposit's $\mathrm{d}W/\mathrm{d}t$ and radiation conventions, and P_{LTH} is one of the nine features every model sees. One predictor is the denominator of the response. A log-linear model can carry the -1 coefficient on $\log P$ exactly, and carry it outside the training range of P without error, because the relation is definitional rather than fitted; a tree approximates it by axis-aligned splits and, by the bound of Sec. 4.1, cannot extrapolate it at all. That admits a reading in which the power law transfers because an identity is exactly log-linear rather than because it has better long-range structure.

Refitting every model on the eight remaining engineering features tests it, and the answer is a real qualification rather than a clean acquittal. The direction survives and the strength does not. The forest still wins cross-validation, by a wider margin than before, 0.198 against 0.316 where it was 0.128 against 0.181, and it still loses under leave-one-label-out. But it loses on **10 of 13** labels rather than 13, and **9 of 11** devices rather than 11, the exact tests move to $p = 0.092$ and $p = 0.065$, from values that were small only under the idealised null of Sec. 2, and the mean paired gap falls from +0.251 to +0.051. C-Mod, JT-60U and JET-ILW change hands, and the gradient booster's mean gap turns negative, so with P removed it beats the power law on average across held-out labels.

A substantial part of the reversal's magnitude therefore depends on P being in the feature set, and the unanimity in particular does. What survives is the direction: under both feature sets the model that wins the interpolation comparison is the one that loses the extrapolation comparison, which is the claim Sec. 4 makes.

The same objection, answered the right way round. Deleting P answers more than the objection asks. P is a real predictor of confinement as well as the target's denominator, and deleting it removes both, which is most of why the ablation above costs so much. The control the objection actually calls for moves W_{th} to the left instead: regress $\log W_{\mathrm{th}}$ on the same nine features, so that P stays among them as an ordinary predictor and only the identity goes.

That control is available. STD5 is a 15-column extract carrying `TAUTH` and `PLTH` and no stored energy, but the full DB5.2.3 revision, already downloaded, pinned by SHA-256 and analysed in Sec. 10, carries `WTH` on 13828 of its 14153 rows, and joining it to STD5 on machine, shot and time recovers a stored energy for **6214** of the 6228 analysed rows. `KAPPAA` is delivered in both files, so the two copies of it are compared across the join and agree to 5×10^{-10}, which is the check that the rows carried across are the rows being analysed. On them $W_{\mathrm{th}}/(\tau_{\mathrm{th}}P)$ has a median of 1.000000 and quartiles of 0.981 and 1.000.

Retargeted, the direction and most of the magnitude survive; the unanimity does not. The forest still wins cross-validation, 0.123 against the power law's 0.193. Under leave-one-label-out it loses on **9 of 13** labels at a mean gap of +0.220, against 13 of 13 and +0.251 on τ_{th}; under leave-one-device-out it loses on **9 of 11** at a mean gap of +0.348, which is larger than the +0.315 the confinement time gives. Across the ITER-size-matched cut it scores **1.081** against the power law's 0.289, so the collapse the paper rests on is untouched. The exact tests move to $p = 0.267$ and $p = 0.065$.

The two controls together separate what the identity carries from what it does not. It carries the unanimity: under either control the count falls from thirteen to nine or ten, and neither exact test then reaches $p = 0.05$. It does not carry the effect. Deleting P, which also deletes a genuine predictor, leaves a mean gap of +0.051; deleting only the identity leaves +0.220 by label and +0.348 by device and leaves the size cut where it was. A reading of this paper as "a log-linear model can represent $\tau = W/P$ exactly and a tree cannot" is therefore available, and this is the measurement that bounds what it can carry. The specification is in `analysis_stored_energy.py` and `results/stored_energy.json`.

Two results this paper leans on do not weaken. The interval collapse of Sec. 7 is unaffected in kind: on a held-out label at nominal 90% the eight-feature intervals cover 42% for the forest and 48% for the booster against 77% for the power law, where the nine-feature figures are 35%, 45% and 83%. Nothing about a conformal interval involves the target's definition, and the measurement agrees. The kernel comparison of Sec. 12 also holds: the bounded squared-exponential rung remains the worst of the three processes, at 0.496 against 0.400 for the linear rung and 0.409 for their sum, where with P it was 0.541 against 0.214 and 0.218. The margin narrows and the ordering does not, and the linear-kernel process still reproduces ridge to four decimals in both arms, which is the control that licenses reading the rest.

The relation is not a defect in the feature set. It is physics, and it is the physics Sec. 8 imposes deliberately: the Connor–Taylor constraint on the exponent of P is derived from $\tau_E = W/P$, and a similarity transformation treating P as an independent scale would produce a constraint no published law satisfies. The constrained rungs therefore cannot be scored under this ablation at all, since the constraint is defined over an exponent vector that includes P.

The estimator this split design is usually paired with. Every comparison above sets a pooled global fit against ensembles, where the clustered-validation literature cited here for the split design [21] pairs it with a third kind of estimator: a mixed model carrying a random intercept per device, from which a device never seen is predicted by the population-level fixed effects alone. It is neither a power law nor a tree and it is the fit matched to the deployment question, so Sec. S10 of the Supplementary Material fits and scores it rather than conceding it. Fitted by restricted maximum likelihood it scores 0.264 per label against the power law's 0.214 and **0.829** per device against 0.212, losing on 10 of the 11 devices, and sweeping its one variance component shows the best value that component could take is the one at which the estimator *is* the pooled power law. The deployment-matched estimator therefore does not displace a pooled power law on an unseen device, and what has been ruled out is narrower than the class: a single random intercept, no random slopes, on eleven groups.

5 A size-extrapolation stress test matched to ITER

Holding out JET still leaves the rest of the database spanning much of its range, so leave-one-out extrapolates in identity while interpolating in size. ITER's major radius is 6.2 m against 3.40 m for the largest row here, a factor of 1.82. That factor is available inside the database: training on the 14 smallest database labels (up to DIII-D, 1.865 m, 3498 rows) and predicting the 4 largest labels (up to JT-60U, 3.40 m, 2730 rows) demands a ratio of 1.823 against ITER's 1.824. The cut is selected by proximity to that ratio rather than by eye, so it is a property of the data.

Two things bound what this cut can carry, and they are why it is a supporting stress test here rather than the result the paper rests on. Its held-out side is 96% JET: of 2730 rows, 1762 are JET and 866 JET-ILW, with JT-60U contributing 100 and TFTR 2. So it is close to a single-device holdout at a size ratio, not a survey of large machines, and the per-label scores below matter more than the pooled one because of it. And the boundary is drawn on major radius alone. Hall *et al.* [24] have argued, in a conference presentation, that the weak size dependence in this database comes from a subset localised in dimensionless space rather than from size as such, which is a different place to cut. Sec. 4.1 measures that rather than conceding it, and the measurement goes their way: across this cut $\log \rho^*$ and $\log \nu^*$ separate as well as size does. This section is therefore transfer across a joint displacement in size, normalised gyroradius and collisionality, not across size alone. The claim the paper rests on is the leave-one-device-out result of Sec. 4, which does not depend on where a size boundary is drawn.

Table 3 places that cut beside the two easier splits. At the cut the power law scores 0.278 against the analytic law's 0.194 and a constant predictor's 1.459. The random forest scores 0.938 and the gradient booster 1.072, so both sit nearer the constant than the power law in absolute error. Asked the question a scaling law exists to answer, the two tree ensembles that lead the conventional cross-validation comparison land closer to a constant than to IPB98(y,2), the law they beat by 36% under cross-validation and which scores 0.194 here. The ordering is identical on all three individually scored held-out labels, so the pooled number is not an artefact of how JET is weighted within the pool.

That control is weaker than it looks, and the composition of the held-out side should be read before the number is. The four labels above the cut are JET (1762 rows), JET-ILW (866), JT-60U (100) and TFTR (2); TFTR falls far below the 30 rows required to report a label on its own and enters only the pooled number. So 2628 of the 2730 held-out rows, 96 % of them, come from a single physical device, and the three individually scored labels are two physical machines rather than three, since JET and JET-ILW are the same tokamak in two wall eras. Seeing the same ordering twice on one machine is not two independent confirmations of it. What this cut establishes is that the tree ensembles collapse across an ITER-sized jump on the rows this database has above one, and what would settle it is a second large device at comparable weight, which this database does not contain. That is a limitation of the question as much as of the split: large machines are few, and their scarcity is what makes the extrapolation hard. This section rests on a large jump where Sec. 4 rests on many machines. Dropping the spherical tokamaks moves the random forest from 0.938 to 0.936, so the effect is not driven by including the spherical devices; other shape or regime covariates that track size are not ruled out by this control.

Table 3: Three questions of increasing difficulty, scored in log-RMSE. The *size cut* column is the ITER-size-matched cut. Reading down it rather than across the table is the point: the two models that win the first column move far toward the constant baseline in the third, landing closer to it than to the power law in absolute error.

Model	discharge CV	LOLO	size cut
IPB98(y,2)	0.199	0.188	**0.194**
ridge, log-linear	0.181	0.214	**0.278**
random forest	0.128	0.465	**0.938**
hist grad. boosting	0.130	0.359	**1.072**
mean baseline	0.869	0.994	1.459

6 A power law with a bounded correction

The diagnosis in Sec. 4.1 suggests a repair that Sec. S9 of the Supplementary Material builds and scores: fit the power law, learn a correction on its log residuals, and damp it, so that what the tree ensemble is bounded on is a residual centred on zero rather than a target that grows with machine size. Across the ITER-size-matched cut a bounded correction takes the power law from 0.278 to **0.206** while an unbounded polynomial correction takes it to 0.356, and the mechanism is measurable: the base law is biased on the held-out machines and the correction points the right way and is largest where the bias is largest.

Two limits keep this a supporting result rather than a recommendation, and both are why it sits in the supplement. Cross-validation selects the most heavily corrected rung in both families while the rung that transfers is the uncorrected one, so a practitioner tuning on the split they would ordinarily use does not arrive at it. And over the whole size sweep the bounded hybrid wins 5 of 8 well-powered cuts rather than all of them, with no rule separating the wins from the losses. Anchoring a learned correction on a frozen power law is also the design of concurrent work on this database [17], and nothing here is a claim on that idea. The controlled version of the comparison, which holds the nonlinear component fixed and varies only the long-range mean, is Sec. 12.1.

7 Calibrated intervals and their collapse

For a next-step device the point error is not the deliverable; the interval is. Split-conformal intervals [25] are calibrated on log residuals, holding back 25% of *discharges* to calibrate.

The construction is specified as follows, so that a reader can reproduce the coverage numbers rather than infer the procedure from them. Within each training fold, the distinct discharge identifiers are permuted with `numpy.random.default_rng` seeded at 20240811, and the first round($0.25\, n_{\text{discharges}}$) of the permutation are the calibration discharges; every row of a calibration discharge is a calibration row, and the model is fitted on the remaining rows only. Arms with fewer than 100 calibration rows are not reported. The conformity score is the absolute log residual $|\log\hat{\tau} - \log\tau|$. The half-width is the $\lceil (n+1)(1-\alpha)\rceil$-th smallest score in ascending order, at $\alpha = 0.10$, which is the finite-sample conformal rank rather than the plain empirical quantile; the $+1$ accounts for the test point itself being one of the $n+1$ exchangeable scores, and where that rank exceeds n the half-width is reported as infinite rather than as a finite but invalid number. Ties need no rule, since the quantile is a rank on a sorted array. The interval is symmetric in $\log\tau$ and is a pointwise interval for a new observation, not for a latent mean. Under grouped cross-validation the per-fold

Table 4: Empirical coverage of nominal 90% split-conformal intervals, and the multiplicative half-width under leave-one-out.

Model	CV	LOLO	ITER cut	width
IPB98(y,2)	90%	89%	88%	1.40×
ridge, log-linear	90%	83%	70%	1.33×
bounded hybrid	90%	64%	76%	1.26×
hist grad. boosting	91%	45%	**0%**	1.23×
random forest	91%	35%	**3%**	1.23×

coverages are pooled over rows; under leave-one-label-out each held-out label is calibrated and scored on its own fold and the coverages are then averaged over machines.

Holding out whole discharges keeps a calibration row and a test row from coming out of the same shot, but the conformity unit is still the row, and rows arrive in correlated clusters of quasi-stationary slices from one discharge. The standard finite-sample guarantee assumes exchangeable conformity scores and clustered rows do not supply that, so every coverage number below is empirical row-level coverage rather than a guaranteed rate. A discharge-level conformity score, or a cluster-aware conformal method of the jackknife+ and CV+ family [43], would be the way to recover a guarantee, and neither is used here.

Interval estimation for a next-step device on this database family is not new: Kardaun [39] constructed confidence and prediction intervals for ITER's confinement time from the predecessor database, by a parametric route rather than a distribution-free one. What is asked here is narrower, and is about a procedure rather than about ITER: whether an interval calibrated the conventional way still covers once the held-out unit is a device.

That proviso is what the section measures. Empirical coverage is near nominal under grouped CV, where every model lands within a point of 90%, and that control is what licenses reading the rest of Table 4. It is a useful in-distribution control rather than a verified guarantee: clustered row-level conformity scores do not satisfy the finite-sample exchangeability condition either, so what the CV column shows is that the construction behaves as intended when the test rows come from labels the model has seen. The exchangeability assumption is no longer justified when the test rows come from a systematically held-out label absent from the model's training and calibration distribution, and what that costs empirically is measured below. That exchangeability is the binding assumption, and that repairing it requires knowing the shift, is the subject of the weighted-conformal literature [26], and calibrated uncertainty degrading under dataset shift is a general finding rather than a fusion one [27]. The random forest's 90% interval then covers 35% of rows on an unseen label and 3% across the ITER-size-matched cut; the gradient booster covers **8** of the 2730 rows there, which is 0.3%.

A coverage that close to zero is worth checking against arithmetic before it is read as a finding, because a calibration bug would produce the same number. The booster's half-width across that cut is 0.221 in logs, calibrated on the rows below it, and its per-label errors above it run from 0.66 to 1.13. A typical residual is therefore three to five times the half-width, and an interval of that size has no opportunity to contain the point. The coverage is not a separate fact from the point errors of Table 3 but the same failure read on a different scale, and the two agree.

The widths barely move: no model's interval changes width by more than 1.5% between the two arms. That is forced rather than found, and it is worth being explicit about which part of it is a measurement. The half-width is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest calibration score, calibration rows are drawn the same way in both arms, and a rank statistic of those scores has no route by which the test distribution could reach it. So the width was never going to move, and the residual 1.5% is fold-to-fold noise in which rows land in calibration. What is measured is the coverage, and what the fixed width means for a reader is the point: the interval carries no signal that it has stopped covering, so it cannot be inspected for the failure it is exhibiting. Figure 2 shows both halves of that: the coverage of every model under each split against the nominal line, and the per-label coverage against extrapolation distance.

Coverage collapses along the same axis the errors grow, but only for the trees: $\rho = -0.77$ for the random forest and -0.54 for the gradient booster against $+0.27$ for the power law, mirroring the per-label error correlations of Sec. 4.1 with the sign flipped. The power law's intervals are somewhat too narrow everywhere rather than catastrophically too narrow far away, and for a next-step device that distinction is most of what matters, because distance is the one thing known in advance. None of this is a defect in conformal prediction. It is a measurement of an assumption being false.

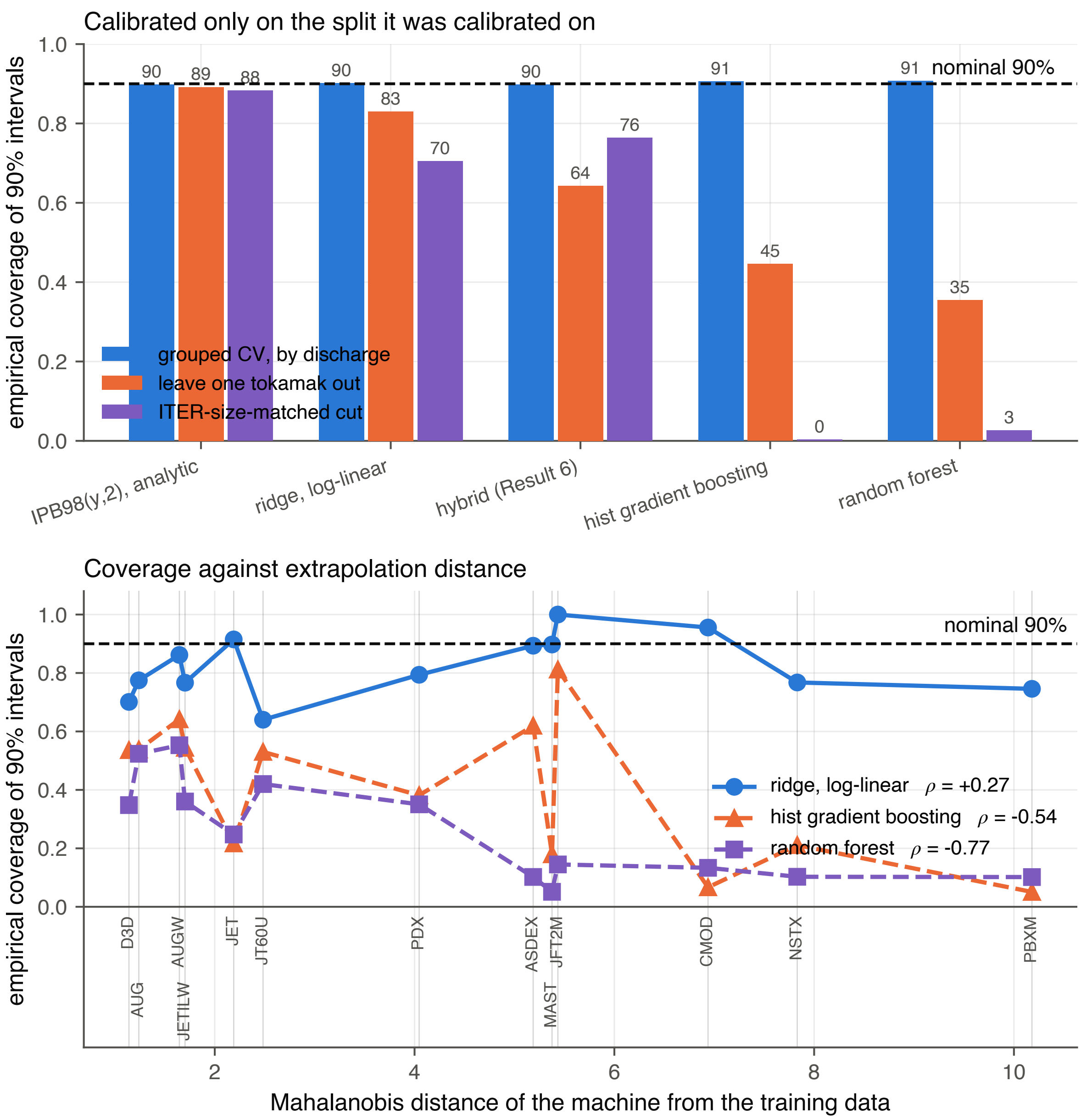

Figure 2: The interval collapse. Top: empirical coverage of nominal 90% split-conformal intervals for each model under the three splits, against the nominal line. Every model is calibrated under grouped cross-validation, and only the linear and constrained fits stay near it once the held-out unit becomes a whole label and then a size range. Bottom: per-label coverage against the Mahalanobis distance of the label from the training data. Under grouped cross-validation coverage is flat on the nominal line at every distance; under leave-one-label-out the tree ensembles fall away with distance and the power law does not. Interval widths behind both panels move by under 1.5% between the splits, so what is drawn is intervals that keep their size and lose their coverage.

8 Dimensional constraints from Connor–Taylor similarity

Every model to this point learns its functional form from the data. None is told any physics. That is worth revisiting, because the field has known since Connor and Taylor [28] that a confinement scaling law is not free: requiring it to be expressible in the dimensionless plasma variables imposes *linear equality constraints on the exponents*. The experimental programme built on that identity is long-standing, and the constraints used here are the same ones it works in: dimensionless-parameter scaling experiments hold the dimensionless groups fixed and vary the remaining scale, which is the transformation family derived below, run on a machine [40, 41]. What is new here is not the constraint but where it is imposed, as an equality on a fitted surface scored out of distribution. A physics assumption is therefore a matrix C, and the fit becomes

$$\min_b \|Xb - y\|^2 \quad \text{subject to} \quad Cb = d, \tag{2}$$

which the Karush–Kuhn–Tucker (KKT) system solves in closed form. Nothing new is needed to run the experiment: the solver is the one written by hand for Sec. 3.1.

The constraints are derived in code from the definitions of ρ^*, β and ν^* rather than transcribing exponents from a paper, which makes the derivation checkable against something external. Written as exponents on (length, field, temperature, density),

$$\rho^* \sim T^{1/2} B^{-1} L^{-1}, \qquad \beta \sim n\, T\, B^{-2}, \qquad \nu^* \sim n\, L\, T^{-2}, \tag{3}$$

and a law expressible in whichever of these a model holds fixed must be invariant under every scale transformation that leaves those groups unchanged. Each such transformation contributes one linear equation in the exponents, so the admissible set is a surface and the number of constraints is set by how many groups are fixed, as Table 5 sets out.

Table 5: The Connor–Taylor hierarchy. Each row fixes a set of dimensionless groups and contributes one linear equation in the exponents per admissible scale transformation, so fixing fewer groups imposes more constraints.

constraint	groups held fixed	rows of C
Kadomtsev [29]	ρ^*, β, ν^*	1
collisionless	ρ^*, β	2
electrostatic	ρ^*	3

Fixing fewer groups admits more transformations and therefore imposes more constraints, which is why the hierarchy runs the way it does. C and d are built by `dimensional.constraint_matrix` from Eq. (3) alone, and their numerical rows are in `results/dimensional.json`.

Constraining a confinement scaling this way is not new, and IPB98(y,2) is itself an instance of it: the high-β Kadomtsev constraint was enforced when it was fitted, alongside a magnetic-field exponent fixed at 0.15 [1, 11]. Its exponents accordingly land on the Kadomtsev surface [29] derived here at a distance of 0.00096, and on the collisionless surface at 0.0045. The first of those is expected rather than surprising, and it is used only as a check that the surfaces derived in code from Eq. (3) agree with the constraint the published law was fitted under. What is new here is not the idea of constraining the exponents but the comparison: a hierarchy of such surfaces, fitted and scored under a validation protocol matched to the deployment problem.

Where the newer published laws sit on these surfaces. That check invites a harder one, which the same machinery runs and which this section should not decline to report about its own subject. The two laws of Sec. 4.2 that transfer best across the ITER-size-matched cut are also the two furthest from every surface derived here. Their triangularity term is dimensionless and contributes no column to C, so the eight exponents the constraint is written over compare directly.

Table 6: Distance from each Connor–Taylor surface, as $\|Cb-d\|$ over the eight exponents the constraint is written over, beside the score each set of exponents gives across the ITER-size-matched cut. The two published laws fitted to this database revision are the two furthest from every surface and the two best across the cut. The numerical rows are in `results/dimensional.json`.

exponents	Kadomtsev	collisionless	electrostatic	ITER cut
IPB98(y,2), published	**0.00096**	0.0045	0.501	0.194
free refit (this database)	0.0031	0.064	0.817	0.279
ITPA20, published	**0.0147**	0.089	0.745	**0.177**
ITPA20-IL, published	**0.0142**	0.106	0.853	**0.165**

Proximity to a surface therefore does not order these four, and no claim here rests on reading it that way. ITPA20 sits fifteen times further from the Kadomtsev surface than IPB98(y,2) and scores 0.177 against 0.194 at the cut, and the free refit sits closer to that surface than either ITPA20 law while scoring 0.279, the worst of the four. A small residual is not a predictor of transfer on this evidence.

What the hierarchy measures is narrower than that, and it is measured the only way a constraint can be: against a refit on these rows where the constraint is the single difference, the same nine features, the same closed-form solver and no hyperparameter separating 0.279 from 0.183. Table 6 is about four different fitting populations and cannot weaken a comparison internal to one.

It does point at what else a fitted law has available, and that is this paper's own argument arriving from the fitting side rather than the scoring side. ITPA20-IL is fitted to an ITER-like selection out of DB5.2.3 rather than to the whole of it, which is what its name denotes. Sec. 1 asks that a validation split match the deployment being claimed; restricting the fitting population to the regime being extrapolated *to* is the same principle applied to the fit rather than to the score, and across this cut it is worth more than any constraint tested here. The two are not competing repairs. A constraint is what a fit can carry when the population cannot be restricted, which is the situation of anyone refitting the published database as this paper does; the ITER-like selection is a choice the ITPA authors could make and this analysis could not, since the machines defining that regime are the ones held out. Both supply structure that the rows alone do not, which is the section's claim, and neither is evidence that the other is unnecessary.

The three rungs do not carry the same evidential status, and separating them matters more than ranking them does. The Kadomtsev rung is *prospective*. It was specified by Kadomtsev in 1975 [29], and enforced when IPB98(y,2) was fitted in 1998 [11], both without reference to anything measured here; imposing it is a prediction this analysis inherited rather than a surface it selected. The collisionless rung is the best of the three at the size cut and was selected on that basis; what the selection costs the claim is set out where the number is reported. The prospective one comes first below.

Take the prospective rung first. The Kadomtsev constraint is *free*: at 0.1808 in sample it matches the unconstrained fit to four decimals, because the data already satisfies it

Table 7: Fitting under the Connor–Taylor constraint hierarchy, ordered by score at the ITER-size-matched cut. The only difference between the first and last rows is the constraint: same features, same solver, no hyperparameter. The Kadomtsev row is the prospective one, fixed by the literature before this analysis; the collisionless row was selected on the scores in this table. LODO is the same held-out score over the 11 physical devices rather than the 13 database labels, so it is the arm Sec. 4.2 argues the paper rests on, applied here to the models this paper recommends rather than only to the ones it criticises.

Model	in sample	CV	LOLO	LODO	ITER cut
power law, collisionless	0.1818	0.182	0.206	0.203	**0.183**
IPB98(y,2), analytic*	n/a	0.199	0.188	0.184	0.194
bounded hybrid (Sec. 6)	n/a	0.151	0.246	0.246	0.206
power law, electrostatic	0.2245	0.225	0.241	0.226	0.237
power law, Kadomtsev	0.1808	0.181	0.210	0.211	0.254
power law, unconstrained	0.1808	0.181	0.214	0.212	0.279

*historical reference, fitted to the DB2.8 predecessor of this database [11] and not refitted within these folds. The unconstrained row is the closed-form least-squares fit this section's solver produces, at 0.2790 across the size cut; the ridge spelling of the same power law reported elsewhere gives 0.2778, and Sec. 3 shows the choice between them carries nothing.

unaided. It nonetheless beats that unconstrained fit at every one of the 15 cuts in the size sweep, including all 8 well-powered ones, where a cut is called well-powered when it trains on at least 1000 rows. Satisfying a constraint on average is not the same as being unable to violate it, and what the constraint buys is a fit that cannot wander along that direction when the training half is small or shifted. The underpowered rungs show it directly: at six training machines the unconstrained fit scores 0.971 and the Kadomtsev fit 0.274. Nothing in that comparison was selected after the fact, which is why it is the result in this section to weigh most heavily.

The collisionless rung is the stronger assumption and the better score. Something can still be said for it that does not depend on the score. It is a physics judgement about the machine being extrapolated *to* rather than a hyperparameter: one adopts it because one expects the target plasma to be collisionless, and ITER is expected to sit at lower normalised collisionality than the devices in this database rather than higher. That points at this rung in advance, for a reason external to the data. The rung also has no within-surface quantity to tune, so there is nothing for a validation split to fit even in principle. Neither observation makes the result prospective, and the disclosure stands as written.

At the ITER-size-matched cut the collisionless power law scores **0.183**, the lowest held-out error of any model fitted here without using the held-out rows (Table 7). That qualifier excludes the published laws, and two of them are lower: ITPA20-IL scores 0.165 there and ITPA20 0.177 (Sec. 4.2), both having been fitted to selections out of the revision analysed here and so having seen the machines above the cut. Nothing below is a claim to have beaten them. The fit is blind to the held-out rows; the *search* is not. No coefficient ever saw a held-out row, but the constraint rung was chosen after seeing how each of the three scored at this cut, so the cut took part in model selection even though it never entered a fit. This is therefore the lowest error among the surfaces examined, not a prospective demonstration that the collisionless surface is the right one. It beats the bounded hybrid of Sec. 6, which was built specifically for that cut, and it beats the analytic law, which was fitted under a Kadomtsev constraint of its own on a predecessor database. Figure 3 shows the trade between what each constraint costs in sample and what it buys at the size cut.

The collisionless constraint costs 0.001 in sample against the unconstrained fit and wins at 8 of 8 well-powered cuts, so it is not a single lucky cut, but it is worse than Kadomtsev at every rung below the well-powered line: it is the stronger assumption and it pays only where there is enough data for constraining to be worth doing. Push one rung further to a beta-independent law and it degrades sharply, to 0.237. There is an optimum in the middle of the hierarchy, and nothing in this analysis predicted where it would fall; as with the hybrid, cross-validation does not select it. Every constrained fit scored slightly worse than the unconstrained one under the cross-validation used here, so CV supplies no signal pointing at the constraint that transfers best. That is an observation about these folds rather than a theorem: a hard equality constraint can only match or worsen the *training* residual, and nothing forces it to worsen a held-out score.

It matters that this is a constraint and not a prior. Supplying the same physics as a Gaussian prior instead, shrunk along the weak singular direction of Sec. 3.1, is worth much less. That comparison sweeps three shrinkage geometries (isotropic, spectral and truncated) over the same nine penalty strengths $10^{-3}, 10^{-2}, \ldots, 10^{5}$, tabulated in full in `results/dimensional_prior_sweep.csv`. Targeting the shrinkage beats isotropic shrinkage at every one of those matched penalty strengths, which is a real effect, but nothing in that family beats simply adopting IPB98's published exponents outright. The hard constraint used here restricts the fit to a surface the Connor–Taylor similarity assumptions permit, where the Gaussian prior tested here favours a location and a direction in parameter space, and in this experiment the surface transfers better. That is a statement about these two families rather than about priors in general: a prior can favour a subspace, and approaches a hard constraint as its variance goes to zero.

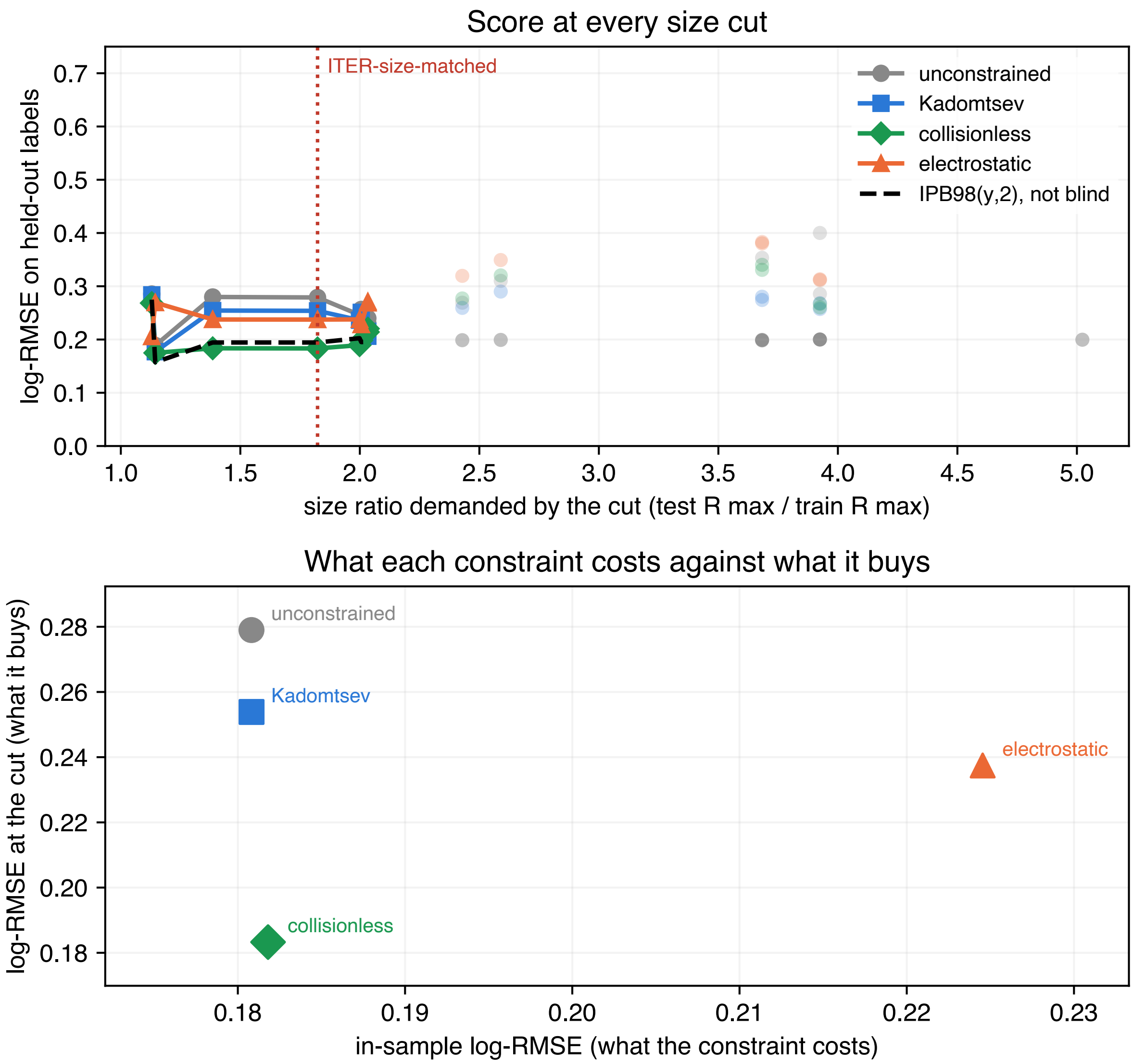

Figure 3: Cost against benefit across the size sweep. Top: the score at every size cut for the constrained and unconstrained fits, with underpowered cuts drawn faint and excluded from claims. Bottom: what each constraint costs in sample against what it buys at the ITER-size-matched cut. The constraint costs almost nothing where the data is plentiful and pays where it is not, and more physics is not monotonically better: the hierarchy has an optimum in its middle, at the collisionless rung, which was itself selected on these scores.

9 Repairing the intervals, and the limit of the repair

The diagnosis in Sec. 7 makes a testable prediction. If the collapse is about exchangeability rather than about the models, then calibrating on held-out *labels* instead of held-out discharges should repair it, and should repair it only as far as that unit reaches. Both halves hold. On a held-out label, empirical row-level coverage returns to within two points of nominal for every model, the random forest going from 35% to 88%. Across the ITER-size-matched cut none of the schemes tested restores it, because every calibration unit is smaller than every test unit, and scaling the interval by extrapolation distance recovers only part of the gap, 3% to 40% for the forest.

Repeating the whole procedure over the 11 physical devices, which removes the overlap that leaves JET in the calibration set when JET-ILW is the target, shows the repair is less complete than the label-level numbers suggest: the booster reaches 77% by device against 90% by label, and the trees buy their coverage at 2.3 to 2.8× multiplicative half-width where the linear models stay near 1.4×. Among the models refitted within folds, the constrained power law of Sec. 8 is the only one whose plain split-conformal coverage stays near nominal at the size cut, at 91%, and its intervals hold there under every scheme.

Section S1 of the Supplementary Material gives both calibration schemes in full, the two coverage tables, and the reasons neither scheme is a finite-sample-valid group-conformal procedure.

10 Robustness on rows the standard analysis set excludes

Everything above rests on one file, which is a central limitation of the preceding analysis. The same OSF project publishes the full DB5.2.3 revision, 14153 rows against STD5's 6228, pinned by its own SHA-256. Matching on (tokamak, shot, time) shows STD5 is an ELMy-H-mode quality selection out of it, and leaves two populations this study had not analysed: 5358 H-mode rows STD5 does not contain, across 12 labels with zero row overlap, and 3860 ohmic and L-mode rows in a different confinement regime, scored against ITER89-P [30] because an H-mode law is the wrong baseline for an L-mode plasma.

The ranking inverts in both arms, and it survives dropping the 336 discharges the disjoint H-mode arm shares with STD5 at different time slices. Counting label–model pairs where a tree ensemble loses to the published law on an unseen label gives 19 of 24 and 10 of 10. Section S2 of the Supplementary Material reports both arms in full, with the positive and negative checks on the row matcher, whose silent failure would otherwise reproduce Sec. 4 by construction. This is emphatically *not* an independent database: both arms come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone.

11 A locked forecast, and a direction that is not size

Everything above is retrospective. To make the disagreement checkable rather than merely described, what each model predicts for three real devices, SPARC, JT-60SA and ITER, is recorded with intervals, a lock date of 31 August 2026 and a content digest over the input rows, so that a later edit to them is visible. The parameter sets are published design values, and none of the three has a *measured* H-mode thermal confinement time at the operating point tabulated, which is what makes these predictions rather than retrodictions. Section S3 of the Supplementary Material gives the table, the interval construction, the disclosures attaching to it and the falsification conditions; the locked values are in `results/forecast.json`. It is a record rather than a validation, and only JT-60SA is checkable in the near term.

On JT-60SA, which sits inside the database's size range and is already operating, all five models agree to within 15%. On ITER, 1.82× beyond it, they disagree by a factor of **8.3**, and for the random forest that gap cannot be closed by tuning: its prediction cannot exceed the largest training target, 1.321 s.

The disagreement among the models that *can* reach ITER is the one a design study would act on, and quoting it as a range hides it. The constrained power law of Sec. 8 predicts 2.837 s at the $Q = 10$ inductive point and the unconstrained refit 2.858 s, against IPB98(y,2)'s 3.591 s: the two fits made here are **21%** below the reference law at the operating point it was used to size. That is the only number in this paper a next-step design would consume, and three things bound how much weight it can take. It is a point estimate from a fit selected on transfer rather than on physics; the constrained rung it comes from was chosen after the size-cut scores had been seen, so its standing is the one Sec. 8 gives it; and it points the same way as the newer published laws, which weaken the size dependence relative to IPB98(y,2) and which Sec. 4.2 finds transferring better here than either fit. A lower projection than IPB98(y,2)'s is therefore what several independent routes agree on, and none of them is this paper's to certify.

One feature of the exercise was not anticipated by anything above. SPARC sits *further* from the training data than ITER does, at a Mahalanobis distance of 6.9 against 4.7, despite being the smaller machine: its 12.2 T field is 2.1 times the largest toroidal field anywhere in the cleaned STD5 rows, 5.82 T on Alcator C-Mod, and every other label tops out at or below 4.80 T. Size is not the only direction in which a next-step device leaves the data behind, and an extrapolation audit organised around size alone would have missed it.

12 Flexibility, or long-range saturation?

Section 4.1 left two readings open, and every model scored to this point confounds them. Flexibility is how much structure a model can learn beyond a power law; *saturation* is whether its predictions stop tracking the features once the input leaves the training support, either because they cannot leave a fixed range or because they revert to a fixed value. The forest is the strict case, provably confined to $[\min y_{\text{train}}, \max y_{\text{train}}]$; the booster never left that range on any split scored here; a squared-exponential Gaussian process reverts to its prior mean. What the three share is the absence of a trend that continues outside the data. The ensembles are flexible and saturating, ridge is neither, and the degree-3 polynomial is flexible and non-saturating but divergent, which is why its worst label is 4.601. Nothing scored so far is flexible, non-saturating and well behaved at long range at once.

A Gaussian process (GP) [31] supplies one, and the property is a choice of kernel rather than a tuning parameter. Section S6 of the Supplementary Material fits three over the same log features, with the same optimiser on the same rows, differing in whether the kernel admits a trend that continues at long range, and the ordering is the one saturation predicts. The squared-exponential rung alone scores 1.948 at the ITER-size-matched cut, worse than a constant predictor, its error tracking distance at $\rho = +0.65$. Adding a dot-product term gives the best cross-validated score in this work, 0.112 against the forest's 0.128, together with 0.191 at that cut and an error flat against distance at $\rho = -0.01$. Its

Table 8: One RBF covariance family and fitting procedure, three mean functions. The constant-mean row reproduces the squared-exponential rung of Sec. S6 of the Supplementary Material to six decimals, which is the control: the decomposition is the same model rewritten. Only the first row saturates. The other two both keep trending, and they do not agree, which is what makes this an ablation rather than an identity. The LODO column repeats the held-out score over the 11 physical devices, where the three separate more widely than they do over the 13 labels.

$m(x)$ in Eq. (4)	CV	held-out	LODO	ITER cut
constant	0.142	0.541	0.693	1.948
fitted power law	0.115	0.212	0.212	0.278
IPB98(y,2), unfitted	**0.112**	**0.189**	**0.186**	**0.186**

intervals behave differently from the calibrated ones as well, which is taken up below.

That ladder suggests the mechanism without isolating it, because a dot-product term enlarges the function class as well as changing the asymptote. The rest of this section removes that objection.

12.1 The same experiment with the nonlinear component specified identically

Write the model as a mean function plus a residual process,

$$\log \tau = m(x) + g_{\mathrm{RBF}}(x), \tag{4}$$

and specify g identically in every arm: same RBF covariance family, same tuning subsample, same seed, same optimiser. Only m changes. The fitted processes are not the same function, because each is conditioned on its own mean's residuals and each re-optimises its hyperparameters on those. What is held fixed is the nonlinear capacity available to the model and the procedure that fits it, so no part of the difference below can come from one arm having more of either.

Two arms would not be enough. An RBF residual decays to zero over a finite length scale, so far outside the training support the prediction *is* the mean function and its score *is* the mean function's score; comparing a constant mean against a fitted power law therefore cannot separate "saturation is the mechanism" from "whatever mean you supply is what you get out here", and only the second would be demonstrated. The third arm separates them. Its mean also keeps trending, and its slope is IPB98(y,2)'s, fixed in 1998 and fitted to nothing here, so two of the three arms have a trend and differ only in which trend it is.

Table 8 reports the three arms. Take the control first: with a constant mean, Eq. (4) returns 0.141533, 0.540967 and 1.947535, which is the RBF row of that rung to every digit computed. Nothing has changed except how the model is written. Giving the same process a power-law mean then moves the ITER-size-matched cut from 1.948 to 0.278, a factor of 7.0, and improves the cross-validated score as well, 0.115 against 0.142. Because the residual GP has the same covariance family, the same capacity and the same fitting procedure in both rows, the difference cannot be attributed to adding nonlinear model capacity.

The second row's cut score is worth reading closely: 0.277823 is plain ridge's score at that cut, to six decimals. Far outside the data the RBF residual has reverted to its training mean, which for a ridge mean is zero, and the model falls back to its mean function. A saturating model reverts to something; the failure of the squared-exponential rung is that a constant mean makes that something a constant, and the repair is not to make the model less flexible but to give it something worth reverting to.

The third row is what stops that being a tautology. If the only content of this section were that the model returns its mean function far from the data, then any trending mean would do equally well and the two trending rows would land together. They do not. Swapping the fitted power law for IPB98(y,2)'s published exponents, with nothing else changed, moves the cut from 0.278 to **0.186**, improves the held-out label score from 0.212 to 0.189 and the cross-validated score from 0.115 to 0.112. Having a trend is necessary and it is not sufficient: which trend decides the answer, and out here the answer is the trend and nothing else. That the best of the three is a law fitted in 1998 to a predecessor database, on exponents this study did not choose, is the same observation Sec. 4.2 makes about the published laws, arriving from a different direction.

Three numbers in this paper answer to the description "a linear trend with a bounded correction", and they differ, so it is worth saying how. The two trending rows above fix the trend first, by penalised least squares in one case and by adopting published exponents in the other, and give the residual process no say in it; they score 0.278 and 0.186 at the cut. The linear-plus-RBF kernel of Sec. 12 instead fits both components jointly by maximising the marginal likelihood, and scores 0.191. None of the three is a disagreement about the mechanism, which all of them exhibit. What separates them is where the trend comes from, and on this database the trend that transfers best came from outside the fit entirely. The kernel is marginally the best cross-validated model here, at 0.11179 against the IPB98-mean arm's 0.11196, which is a tie at any precision worth quoting; out of distribution the two part company, 0.218 against 0.189 on a held-out label. None of the three displaces the constrained fit of Sec. 8 at 0.183, and Sec. 4.2 reports two published laws below that.

Across the model families tested here, this resolves the choice Sec. 4.1 left open: extrapolation failure tracks long-range saturation rather than flexibility alone. The set of models in Sec. 4 is measured correctly, and every flexible member of it was either bounded or divergent, so nothing in it could have separated the two properties; that is why the reading there was flagged as provisional rather than asserted. "Do not use a flexible model out of distribution" is the wrong lesson. "Do not use a model whose predictions stop tracking the features once the input leaves the training data" is the right one, and the random forest's training-target ceiling is its strictest case. This does not displace the constrained power law, which still wins the ITER-size-matched cut with no within-surface hyperparameter to tune, and the process does not beat the power law on a held-out label (0.218 against 0.214). Held out one physical device at a time, which is the arm Sec. 4.2 argues for, it scores 0.219 against the power law's 0.211 and is worse on 5 of the 11. So the process ties the power law under the stricter split as it does under the looser one, and the statement that it does not reverse is made on the split this paper says the claim belongs to rather than only on the finer partition.

A Gaussian process also supplies an interval by construction rather than by calibration, so Sec. 7's question can be put to a model that was never calibrated. At the ITER-size-matched cut, at nominal 90%, the flexible unbounded rung covers 92.5%, where split-conformal intervals give the random forest 3% and the gradient booster none. It is the only flexible model tested whose *native* posterior intervals reach nominal coverage out of distribution without conformal recalibration. That is a statement about where the interval comes from rather than about coverage alone: the constrained power law of Sec. 8 also holds near nominal at this cut (91%, Sec. 9), but through an externally calibrated split-conformal interval rather than one the model supplies itself. The bounded rung is the instructive failure: it is the one model that does become vague out of distribution, with a half-width four times any other, and it still covers 16.7%, having reverted to a prior mean nowhere near the answer. Width was never the problem.

13 Discussion

The practical consequence of Table 1 is narrow and specific. It is not that machine learning has nothing to offer confinement scaling, and not that the models scored here are badly built. It is that conventional within-database evaluations do not measure the quantity the law is used for. Cross-validation grouped by discharge, or by shot, or by time slice, holds out rows from machines that remain in the training set. It measures interpolation within known devices. A scaling law is used to size a device that does not exist, which is extrapolation to an unknown one, and on this database the two comparisons do not merely differ in magnitude: they order the candidate models in opposite directions. A gain reported under the first is not evidence about the second.

This is the selection argument of Sec. 1, now measured, and it has to be stated at the width the evidence carries. Over the three primary models scored here, all of which saturate, the convention does not merely fail to inform about extrapolation: it picks the worst of the three extrapolators on offer, and the more carefully it is followed the more confidently it does so. How far that generalises depends on how much of the literature on this database also fits saturating models, and this paper does not measure that. Tree ensembles are fitted to it [20]; whether a variational or sampling-based Bayesian neural network [19], or a deep ensemble under feature alignment [18], saturates is a question about those architectures that only scoring them under this split would settle, and a network with a linear output layer would not. None of the four is scored here and no claim is made about any of them. It is not a property of cross-validation as such, and this paper holds its own counterexample. Enlarge the candidate set with a model whose predictions keep trending and the selection rule repairs itself. The linear-plus-RBF Gaussian process of Sec. 12 is the best cross-validated model in this work, at 0.112 against the random forest's 0.128, and it is the flexible model that does not collapse: 0.218 on a held-out label, which ties the power law's 0.214 rather than beating it, and 0.191 across the ITER-size-matched cut against 0.278, with a per-label error flat against distance ($\rho = -0.01$). A practitioner who put that model in the comparison would be selected onto it. What fails is therefore not the score but the set of models the score is applied to, and the property separating the survivors from the casualties is long-range saturation, which Sec. 12.1 isolates and which can be read off an estimator before any device is held out.

13.1 What this changes in practice

The consequence is primarily a reporting convention rather than a modelling one, and it is short enough to state as a checklist. For any confinement model offered as an alternative to a power law on this database:

1. Report leave-one-device-out beside cross-validation. It costs one refit per device, on the same rows and the same features, and the two numbers print together. Where they agree the interpolation number is informative. Where they disagree, as here by a factor of 3.6, the disagreement is the result, and the interpolation number should not be quoted on its own.

2. Report how far the target operating point sits from the training distribution. This

needs no held-out device. The Mahalanobis distance of a label's mean log-feature vector tracked per-label error at $\rho = +0.85$ for the random forest and at $\rho = -0.06$ for the power law, so it separates the models that will degrade from the ones that will not, without waiting for the machine to exist.

3. Say whether the model's predictions stop tracking the features outside the training support, and separately whether they can leave the range of the training targets at all. Both are properties of the estimator, checkable before anything is fitted. The second is the sharper statement and the narrower one: a random forest provably cannot leave that range, since every prediction is an average of training targets, so it cannot project a machine expected to exceed the largest confinement time it has seen. On this database that bound is active on one held-out label of the thirteen and across the size cut, and the forest loses worst on labels where it is nowhere near binding, so it is the first of the two properties that carries the failure here.

4. Do not carry a calibrated interval across a device boundary without saying so, and do not read its width as evidence about that. Split conformal assumes exchangeable conformity scores, and across a held-out device the assumption is false. What it costs is not a wider interval but the same interval with the coverage removed: 90% nominal delivering 35% on an unseen label and 3% across the size cut, at a width that moves by under 1.5% because a half-width fixed by the calibration rank cannot move. The interval is therefore the wrong place to look for the warning, and the distance of item 2 is the right one.

Only the first needs a refit. The other three cost nothing and are computable from the fitted model and the intended operating point alone.

The positive findings point the same way, and both are about the far field rather than about capacity. Constraining the exponents to a Connor–Taylor surface and adding a linear component to a Gaussian process kernel are unrelated operations, and both supply long-range structure. They are not equivalent in what else they change: the constraint removes freedom from the fit, while the linear kernel also enlarges the GP's function class, so that rung on its own cannot separate the asymptote from the capacity. The mean-function ablation of Sec. 12.1 is the cleaner test, since there the nonlinear covariance family and the procedure fitting it are held fixed and only the long-range mean changes. It should still be read as a statement about the families tested, not about flexible models in general.

For practice, the cheaper option was also the stronger one out of distribution. A power law refitted on the same rows is a stronger baseline out of distribution than any tree ensemble scored here, and it costs a single least-squares solve; the estimator hardly matters, since ordinary least squares on the independent columns reproduces the reported ridge to within 0.0012 on every split. Where a flexible model is wanted, the evidence here favours one that keeps trending, and the linear-plus-RBF Gaussian process was both the best cross-validated model in this study, at 0.112, and the only flexible model whose native posterior intervals reached nominal coverage out of distribution without conformal recalibration; the constrained power law reaches comparable coverage there through split conformal instead. Where a physics constraint is available, imposing it is close to free and helps out of distribution. The rung the literature fixed in advance costs nothing in sample and beats the unconstrained fit at all 15 cuts of the size sweep; the other costs 0.001 and was the best-scoring blind fit at the size cut, having been selected on it.

None of this is settled by one database, and every fusion result above rests on one ITPA file. Two replications outside fusion are reported in Secs. S4 and S5 of the Supplementary

Material, and the second of them establishes something these rows structurally cannot. Its nested predictor ladder varies how many predictors the models see while holding the rows, the splits and the model specifications fixed, and finds the extrapolation failure at every rung, including the rungs where the flexible model has no interpolation advantage to lose. That is the harder case, and the one with no warning sign at all: a practitioner running the conventional comparison there sees nothing wrong and still cannot extrapolate. HDB5 cannot show it, because on these nine features the flexible model always wins the interpolation split. Running the same audit on the published laws of those two fields, Kleiber's exponent [32] and the allometry of the BAAD compilation [33], reproduces the extrapolation failure in both, while the ranking inversion appears only where the flexible model first wins the interpolation split. Secs. S4 and S5 carry the construction and the caveats.

14 Limitations

What this database can resolve is worth stating once, because several comparisons above are close. Two models whose per-label scores differ by less than about 0.01 in log-RMSE should be read as tied here, which is roughly the spread the bootstrap puts on a single label's score and roughly what a fold assignment moves the cross-validated margin by. On that reading the Gaussian process at 0.218 and the power law at 0.214 are tied, as the text says; ITPA20's 0.191 and IPB98(y,2)'s 0.188 are tied; and the orderings the paper does claim, the forest's 0.465 against 0.214 and the size cut's 0.938 against 0.278, are an order of magnitude clear of it. No ordering finer than 0.01 is asserted anywhere, and none should be read into a table.

The refit population is not IPB98's, and the refit exponents should not be read as a correction to it. Two machines (JET and ASDEX Upgrade with their variants) supply 77% of rows. Only 13 of the 18 database labels are scored under leave-one-label-out; the five excluded are the smallest and most unlike the rest, so the reported gap is if anything optimistic. Thirteen labels is a small bootstrap and the paired differences are the more reliable statistic.

The training-target bound of Sec. 4.1 is a theorem for the random forest and an observation for the gradient booster: boosting sums tree outputs rather than averaging targets, so nothing confines it to the training range, and what is reported here is only that it stayed inside that range on all 14 splits scored. The ITER-size-matched cut reproduces ITER's size ratio and nothing else: ITER is a burning plasma dominated by alpha self-heating at a gain no row here approaches, so clearing this bar is necessary and not sufficient. Its composition is the sharper limit and is restated here because several results rest on it. Of the 2730 held-out rows, 2628 are one physical device, and the three individually scored labels are two machines rather than three. The conformal collapse of Sec. 7, the bounded correction of Sec. 6, the constrained fit's 0.183, the Gaussian process's 0.191 and the mean-function ablation of Sec. 12.1 are all scored across it, so all five are one-machine evidence about a large jump, and the claim that rests on many machines is the leave-one-device-out result of Sec. 4 instead. Nothing in this database fixes that: a second large device at comparable weight is what the comparison needs and what does not exist.

The per-label coverages in Sec. 7 are proportions over as few as 39 rows and carry binomial uncertainty of ten points or more; the pooled contrasts are far larger than that noise, the within-column ordering is not. The constraint hierarchy of Sec. 8 has an optimum in its middle that nothing here predicted and that cross-validation cannot select, so the result is that a constraint helps rather than that this constraint is the right one.

The robustness arms of Sec. 10 are disjoint in rows but not independent in provenance: both come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone. The H-mode arm is disjoint in rows and not, as first constructed, in discharges; Sec. 10 reports the rerun with all 336 shared discharges removed, which the reversal survives. The locked forecast of Sec. 11 is a record, not a validation; only JT-60SA can be checked against measurement in the near term, that check requires an H-mode campaign rather than the completed L-mode one and no date is put on when that arrives, and it is the one device of the three that sits inside the database's size range.

The two biological replications carry their own limitations, stated in full in Secs. S4 and S5 of the Supplementary Material: the metabolic-rate arm has one predictor and eleven groups that are phylogenetically related rather than independent, which the comparative-biology literature handles with phylogenetically controlled regression [34] and this paper does not, and the plant ladder is one ordering of one selectively intersected set of predictors, so it establishes that a crossing exists and that the inversion tracks it, not that three is a threshold. Neither establishes a rate for anything.

Every power-law fit here treats the engineering parameters as measured without error, which the confinement-scaling literature has objected to for as long as these fits have been published, because the predictors carry real measurement uncertainty; Kardaun [39] treats the problem directly, Cordey *et al.* [42] refit the database under it, and Verdoolaege *et al.* [6] give a regression for confinement data built to be robust to exactly the noise an ordinary least-squares fit assumes away. Refitting Eq. (1) by orthogonal distance regression, which spreads the error across the predictors instead, moves the exponents enormously: up to 5.6 on the inverse aspect ratio, with the major radius going from 1.58 to 5.40. It also more than doubles the in-sample error, 0.454 against 0.181. Both are symptoms of one thing rather than a better fit. With no per-variable measurement uncertainties published for this database, an errors-in-variables (EIV) estimator has nothing pinning it down along the weak singular direction of Sec. 3.1, which carries 0.3% of the design's variance, and it runs away along exactly that direction. The honest statement is narrower than the one this paragraph used to make. These are least-squares exponents; an EIV fit would differ; and orthogonal distance regression, which is the naive estimator, is not the tool the field uses on this problem. Kardaun [39] and Cordey *et al.* [42] both worked it on this database family, and Verdoolaege *et al.* [6] give a geodesic least-squares regression built for exactly the noise structure ODR mishandles here. None of the three is attempted in this paper. What is reported is that the naive estimator runs away along the weak direction, which is a fact about the naive estimator, and the question of what a credible measurement-error fit would give is left open rather than answered.

The mixed model of Sec. 4.2 is scored on eleven devices, which is few enough that its one variance component is poorly determined: REML places λ anywhere between 2.0 and 143 depending on which device is held out. The sweep is reported for that reason and is what the conclusion rests on, since it does not depend on the optimiser. It also carries a single random intercept and no random slopes, so what has been ruled out is that a device-level *level* shift is the missing structure, not that some richer hierarchical specification would fail the same way. More broadly, what the results above establish is that the ordering of these models reverses and that nothing scored here displaces the pooled power law on an unseen device, which is not the same as its being the best available predictor of one.

Every fit also weights rows equally, so JET and ASDEX Upgrade set most of what each model learns even though the target is transfer to a device neither resembles. Refitting every model with each label weighted equally, reported in Sec. 4.2, costs the power law more than the forest and takes the win count to 12 of 13. The row imbalance is therefore not the source of the power law's advantage, but it is not irrelevant to its size either, and one weighting is not a characterisation of how the result behaves under reweighting in general.

The Gaussian-process intervals of Sec. 12 are equal-tailed posterior intervals on $\log \tau$ that include the fitted observation-noise term and are exponentiated to τ; that is the closest available analogue of a conformal interval, which is also a statement about a new observation rather than a latent mean, but the two are not the same construction and the comparison should be read as such. Those kernels are tuned by marginal likelihood on a seeded subsample of each training fold, with only the hyperparameters coming from the subsample and the posterior then conditioned on the whole fold, which is stable across the range that was affordable (under 4% in every hyperparameter between 750 and 1500 rows) but is not a proof that the full-data optimum agrees; and "linear plus RBF" is one reading of "a power law with a bounded correction" among several, with an anisotropic kernel, a Matern kernel and a linear part constrained as in Sec. 8 all untested. The last of these is the obvious next experiment, since the constraint and the kernel are the two things that work here and nothing has tried them together.

15 Conclusion

On real confinement data, the model that wins cross-validation by 29% over a log-linear power law refitted inside the same folds loses to that same power law on all 13 held-out database labels and on all 11 physical devices once the JET and ASDEX Upgrade variants are collapsed, and to the published IPB98(y,2) law as well, which is the historical reference of Sec. 1 rather than a fold-fitted competitor.

Those counts are unanimous because the loss power is among the nine features and is also the target's denominator, which a log-linear model represents exactly and a tree cannot. Two controls in Sec. 4.2 separate what that identity carries from what it does not. Regressing the stored energy instead, which leaves the loss power in the feature set as an ordinary predictor and removes only the identity, takes the count to 9 of 13 and 9 of 11 and leaves the magnitude alone: the mean paired gap is +0.220 by label and +0.348 by device, against +0.251 and +0.315, and across the size cut the forest scores 1.081 against the power law's 0.289. Deleting the loss power outright, which also deletes a real predictor, is the harsher experiment and costs more, at 10 of 13 and a mean gap of +0.051. So the identity carries the unanimity and the exact tests, which fall below significance under either control, and it does not carry the effect. Three separate diagnostics point at that failure and none of them is a tuning fix, and the mean-function ablation of Sec. 12.1 identifies the mechanism as long-range saturation; the three diagnostics on their own cannot separate it from flexibility. That ablation is run in the engineering coordinates, where several features are close to device labels, so what it establishes is the mechanism in the coordinates the literature fits in rather than in general; the reversal itself is shown in Sec. 4.2 to survive the change to dimensionless coordinates, at a reduced margin. At the size extrapolation that separates this database from ITER, the two tree ensembles that lead the conventional comparison land closer to a constant predictor than to the power law, and their nominal 90% intervals cover 3% and 0% of the held-out rows at unchanged width. The inversion survives rows the standard set does not contain, a second confinement regime, the collapse of 13 labels into 11 devices, either aggregation of the score, a nested hyperparameter search under two selection procedures, and shuffled fold assignments, so it is not an artefact of a single bookkeeping choice. What the models disagree about is recorded rather than argued: on a machine inside the database's size range they agree to within 15%, and on ITER they differ by a factor of 8.3.

The repair that scored best is not additional model flexibility at all: it is a dimensional constraint on the existing power law, and the two rungs tested carry different weight. The Kadomtsev rung was fixed by the literature in 1975 and enforced when IPB98(y,2) was itself fitted, so imposing it here selected nothing; it costs nothing in sample and beats the unconstrained fit at all 15 cuts of the size sweep. The collisionless rung gave the lowest error of any model fitted here without using the held-out rows at the ITER-size-matched cut, but was selected after those scores had been seen, so it is the lowest error among the surfaces examined rather than a prediction that this surface is the right one. Two published laws fitted to this database revision score lower there still (Sec. 4.2), and they had seen the machines above the cut.

Two results temper the diagnosis and neither weakens the practical advice. In the nested predictor ladder of Sec. S5 the extrapolation failure persists at every predictor count tested while the ranking inversion does not, because the inversion needs the flexible model to win the interpolation split first: a field whose scaling law has one predictor faces the same danger with no warning at all. And separating flexibility from long-range saturation shows the failure belongs to the second. Holding the nonlinear component's specification fixed and varying only the mean function moves the ITER-size-matched cut by a factor of 7.0, and the better arm lands exactly on the power law's score there, because far from the data its correction has reverted to zero and left the trend behind.

The lesson is not that flexible models cannot be trusted beyond their data. It is that a model whose predictions stop tracking the features once the input leaves the training data is structurally unsuited to a problem where the target itself has to leave that range. The two are easy to confuse because the flexible models that win the conventional interpolation comparison here are the saturating ones, while the non-saturating polynomial control becomes unstable in extrapolation instead.

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Two generative AI systems were used in preparing this work, and each is named with its version and its role.

Anthropic Claude Opus 5 (model `claude-opus-5`) assisted with developing, debugging and executing author-specified analysis code in the accompanying repository, and with drafting and editing passages of this manuscript.

OpenAI GPT-5.6 Sol was used to critique complete drafts against journal policy and for scientific and historical accuracy, to support the literature review, and to identify methodological issues and propose robustness checks. Several of its suggestions were adopted after the author judged them worth pursuing. They include the corrected account of how

IPB98(y,2) was fitted, in Sec. 4 and Sec. 8; the recomputation of the distance correlation over the 11 physical devices in Sec. 4.1; the device-level calibration of Sec. 9; and several of the references in Sec. 3.1 and Sec. 11, which the author then located and checked against their published records.

Neither system is an author. The author designed the study, decided which suggestions to pursue, specified and implemented every analysis reported here, verified every reported number against the generated artefacts, checked every reference against its published record, and takes full responsibility for the content, the conclusions and any error in them. No text was incorporated without the author reading and verifying it.

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Competing interests

The author declares no competing interests.

Author contributions

Sole author, in CRediT terms. L. Salcedo: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing (original draft), writing (review and editing), visualization, project administration.

Data availability

Four third-party datasets underlie this work. None is redistributed here; each is fetched from its public source by a script in the repository and verified against a pinned SHA-256 on load, so every number above refers to one specific set of bytes rather than to whatever a host currently serves.

- **ITPA Global H-mode Confinement Database, standard analysis set STD5 (DB5.2.3).** The basis of every fusion result. Openly available from the OSF deposit *International Global H-Mode Confinement Database* [35], described in Ref. [11]. File `hdb5_std5.csv`.
- **The full DB5.2.3 revision.** Used only for the replication of Sec. 10, which scores rows STD5 excludes. It is a separate download from the same OSF project (`https://osf.io/download/zhwa3/`) and is not a superset this paper generated. File `hdb5_db523.csv`.
- **Mammalian basal metabolic rate.** The Figshare deposit of Capellini, Venditti and Barton [36]. Used in the Supplementary Material. File `allometry_bmr.txt`.
- **Biomass And Allometry Database (BAAD) v1.0.1.** Released CC0; the data paper is Ref. [33] and the archive analysed here is release v1.0.1 [37]. Used in the Supplementary Material. File `baad_data.zip`.

Table 9: SHA-256 of each third-party input, verified on load. A file whose bytes differ from these is refused rather than analysed, so every number in this paper refers to one specific set of bytes.

file	SHA-256
`hdb5_std5.csv`	`67601c2da5c51f90cf6298ff499cccc7` `4d09ac80c2b98c7dde0d8db3ebb9ac5b`
`hdb5_db523.csv`	`7a48e34379663e3e298924990f05cd8f` `16b8581516bcfa7bb8f438f83ae80ab6`
`allometry_bmr.txt`	`ab22d9ae8f35e96d7fbf5d8e53053d64` `7f0594d74f8917888169993ce668571e`
`baad_data.zip`	`0375f012475658c039e3dcef398951e8` `f68cdeb5430eec8cd65548965496cfe2`

The pinned fingerprints are those of Table 9.
The locked forecast of Sec. 11 additionally carries a content digest over its input rows, so that a later edit to the operating points is visible:

```
960d537ee3df8780afd2054e51d01a268
f8803aca64a830dba37fd1ff5319192
```

All code, the full writeup with every table and limitation, and the scripts that regenerate every number and figure in this paper and its Supplementary Material are openly available under the MIT licence at `https://github.com/lsalcedo100/FusionFlux`, and the exact version used for this paper is archived with a persistent identifier [38]. The analysis code is pinned as firmly as its inputs: every number and figure above was produced at commit

```
6854867466f08bfdcf902ddec6b59e813639ce16
```

Every script verifies that fingerprint against the pin as it loads the data and refuses to run when the two disagree, so no number above can have been computed from a different revision of the database. Several of the JSON outputs under `results/` also record the digest they verified, which carries that provenance into the artefact itself.

The prose is bound to those artefacts in the same way. **One hundred and fourteen** of the numbers printed above, and **one hundred and forty-four** across this paper and its Supplementary Material, are tied by test to the field of the generated file they came from, in both directions: if an analysis is rerun and its output moves, the rendered value stops matching the printed one and the suite fails; if a number in the text is edited, it stops matching the artefact. A figure that disagrees with the file behind it therefore cannot survive a test run, which is the check a reader would otherwise have to perform by hand.

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